

Complex opioid driven modulation of glutamatergic and cholinergic neurotransmission in a GABAergic brain nucleus associated with emotion, reward and addiction

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Ramesh Chittajallu , Anna Vlachos, Adam P Caccavano, Xiaoqing Q Yuan, Steven Hunt, Daniel Abebe, Edra London, Kenneth A Pelkey, Chris J McBain

Section on Cellular and Synaptic Physiology, Eunice Kennedy Shriver National Institute of Child Health and Human Development, National Institutes of Health, Bethesda, United States

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This study presents **important** information about the role of mu opioid receptors in neurotransmission between the medial habenula and the interpeduncular nucleus. The authors provide **convincing** evidence that mu opioid receptor activation has differential effects on transmission from substance P neurons and cholinergic neurons, and that blockade of potassium channels can unmask a nicotinic cholinergic synaptic response. This work will be of high interest to those studying this brain region, and potentially to the larger neuroscience community studying motivated behavior.

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Abstract

The medial habenula (mHb)/interpeduncular nucleus (IPN) circuitry is resident to divergent molecular, neurochemical and cellular components which, in concert, perform computations to drive emotion, reward and addiction behaviors. Although housing one of the most prominent mu opioid receptor (mOR) expression levels in the brain, remarkably little is known as to how they impact mHb/IPN circuit function at the granular level. In this study, our systematic functional and pharmacogenetic analyses demonstrate that mOR activation attenuates glutamatergic signaling whilst producing an opposing potentiation of glutamatergic/cholinergic co-transmission mediated by mHb substance P and cholinergic neurons, respectively. Intriguingly, this latter non-canonical augmentation is developmentally regulated only emerging during later postnatal stages. In addition, we reveal that specific potassium channels act as a molecular brake on nicotinic receptor signaling in the IPN with the opioid mediated potentiation of this arm of neurotransmission being operational only following attenuation of Kv1 function. Thus, mORs play a complex role in shaping the salience of distinct afferent inputs and transmitter modalities that ultimately influences

synaptic recruitment of downstream GABAergic IPN neurons. Together, these observations provide a framework for future investigations aimed at identifying the neural underpinnings of maladaptive behaviors that can emerge when opioids, including potent synthetic analogs such as fentanyl, modulate or hijack this circuitry during the vulnerable stages of adolescence and in adulthood.

Introduction

Substance use disorders (SUDs) are agnostic to demographic group posing a widespread public health crisis impacting many communities¹. Drugs of misuse influence synaptic function resulting in acute and protracted circuit adaptations that can exacerbate compulsive behaviors resulting in dependence^{2,3}. Mu-opioid receptors (mORs) represent the primary molecular substrate responsible for both the clinical efficacious (i.e. pain management) and the euphoric effects promoting opioid misuse and addiction⁴. Additionally, mainly via liberation of endogenous opioids⁵, mORs also facilitate the addictive propensity of other substances^{5–7} e.g. alcohol and nicotine which, together with opioids, comprise the three most prevalent drugs of abuse worldwide¹. mORs are expressed by varied neural cell-types resident in numerous regions within the brain^{7–9}. Thus, detailed investigations as to how mOR activation impacts circuit dynamics constitutes an important endeavor critical for identification of potential therapeutic interventions aimed at alleviating the deleterious societal consequences of SUDs.

Here, we focus on a relatively understudied brain circuit comprising the medial habenula (mHb) and interpeduncular nucleus (IPN). The mHb is a bilateral epithalamic structure receiving significant synaptic input from, for example, the diagonal band and various septal regions^{10–12}. The efferent output of the mHb is predominantly mediated by two distinct populations, substance P (SP) and cholinergic neurons, that primarily impinge on the IPN^{11,13}. This latter structure in the ventral midbrain is mainly populated with GABAergic neurons that influence activity in downstream brain regions such as the raphe nuclei and ventral tegmental area¹⁴. Thus, the mHb/IPN axis is anatomically positioned to integrate incoming limbic forebrain signals to ultimately control the function of these midbrain monoaminergic nuclei. It is therefore unsurprising that, perturbation/modulation of the synaptic recruitment or activity of IPN GABAergic neurons precipitates varied emotion, reward and addiction phenotypes^{13–22}.

The mHb/IPN axes houses one of the highest densities of mORs in the central nervous system^{23,24} thus representing a prominent brain locus for opioid action. Despite this, descriptions regarding the role of mHb/IPN mORs in shaping behavior are still in their relative infancy^{25,26}. mOR activation and subsequent G-protein signaling (Gi/o) classically produces an acute direct inhibitory effect in many neuronal types typified by membrane potential hyperpolarization via activation of inward rectifying potassium (GIRK) channels and/or reduced function of voltage-gated calcium channels function that are coupled to neurotransmitter release^{8,27}. Strikingly, there is a dearth of information concerning the cellular effects and subsequent circuit influence of mOR signaling in the mHb/IPN rendering the neural correlates underlying the behavioral outcomes elicited by opioids unclear at present. Interestingly, direct excitatory effects of mOR activation are apparent in certain areas of the nervous system either under basal conditions or following chronic drug regimens^{28–31}. Furthermore, within the IPN itself, GABA_B-receptor activation (another Gi-linked receptor) produces an increase in synaptic transmission^{32–34} questioning whether this surprising effect is generalized to other members of this receptor family expressed within this region such as mORs.

The central role of the mHb/IPN circuitry underlying numerous behavioral aspects relating to emotion and the cycle of addiction, the conspicuous expression of mORs and a clear precedent for non-canonical effects of Gi-linked receptors together formed the impetus for the current work.

Employing cell-type conditional optogenetics in combination with slice electrophysiology and pharmacogenetics we systematically dissect the effects of opioids on the mHb/IPN circuitry with a particular focus on synaptic signaling between these structures.

Results

The two major neuronal populations of the mHb, substance P (SP) and cholinergic subtypes can be distinguished by virtue of *Tac1* and *ChAT* gene expression. Leveraging the Allen Brain Institute's publicly available whole mouse brain 10X single cell RNAseq dataset (https://alleninstitute.github.io/abc_atlas_access/intro.html)³⁵ it is evident that these two neuronal classes within the mHb cluster (Subclass 145 MH) can be parsed based on their transcriptomic profiles (**Fig. 1a**). Spatially, the SP and cholinergic neurons largely segregate to the dorsal versus ventral mHb, respectively (**Fig. 1b,c**). The projection pattern as delineated by use of specific Cre transgenic mouse lines (*ChATCre* and *Tac1Cre*) when crossed to a conditional TdTomato reporter (*Ai9*), reveals a predominant innervation of IPL by SP neurons whereas cholinergic neurons impinge on the IPR/IPC subdivisions (**Fig. 1c**). Together these distinct neuronal classes provide much of the afferent input to the IPN via the fasciculus retroflexus (fr) axonal tracts (**Fig. 1c**) ultimately dictating IPN neuronal recruitment. At the mRNA level, two of the designated three *ChAT* neuronal superclusters and the single *Tac1* supercluster clearly demonstrate significant expression of *OPRM1* (mean expression levels including zero values = 0.33, 6.8 and 4.1 in *ChAT* Superclusters 0632, 0634, 0635, respectively and 8.7 in the *Tac1* Supercluster 0633; **Fig. 1d**). At the protein level, and as previously reported²³, we confirm that mOR expression is notable in the mHb, the fasciculus retroflexus axonal tracts and particularly prevalent in the IPN (**Fig. 1e**). Within this latter structure, high resolution microscopy clearly show mOR expression is densest in the lateral IPN (IPL) with intermediate and relatively lower levels found in the rostral IPN (IPR) and central IPN (IPC) subregions, respectively (**Fig. 1e,f**). Selective stimulation of each of these mHb neuronal populations in isolation is achieved using Cre-mediated conditional expression of channel rhodopsin by crossing *Ai32* mice with either *ChATCre* or *TAC1Cre* mice; see methods for details). Adopting this optogenetic approach we investigate how mOR receptor activation modulates habenulo-interpeduncular synaptic dialogue imparted by these parallel yet distinct afferent systems in adult mice ($p > 40$). Both SP and cholinergic neurons express the glutamate vesicular transporter, *VGLUT1* (*Slc17a7*; **Fig. 1d**) and competently release this excitatory neurotransmitter^{21,36,37}.

mOR activation reduces substance P neuronal mediated glutamatergic transmission onto IPL GABAergic neurons

Since the highest expression of mOR was found in the IPL (**Fig. 1e,f**), we initially probed the effect of a selective mOR agonist (DAMGO, 500 nM) on AMPA receptor mediated transmission mediated by SP neurons that prominently innervate this subregion (**Fig. 1c**). DAMGO application results in a significant decrease in light-driven (470 nm; typically, 10-50 % arbitrary LED power equating to approximately 0.4 – 3.4 mW/mm²; CoolLED illumination system) AMPAR mediated EPSCs in *TAC1Cre:Ai32* mice that is partially reversed upon washout of the agonist (**Fig. 2a,b**). This is accompanied by an increase in S2/S1 paired pulse ratio (PPR) suggesting the modulation is driven by presynaptic changes in release probability (**Fig. 2c**). Thus, neurotransmitter release by this specific afferent input to the IPL is directly, negatively modulated like that seen in many other neural circuits within the brain⁸.

mOR activation augments the strength and fidelity of glutamatergic transmission by cholinergic neurons resulting in

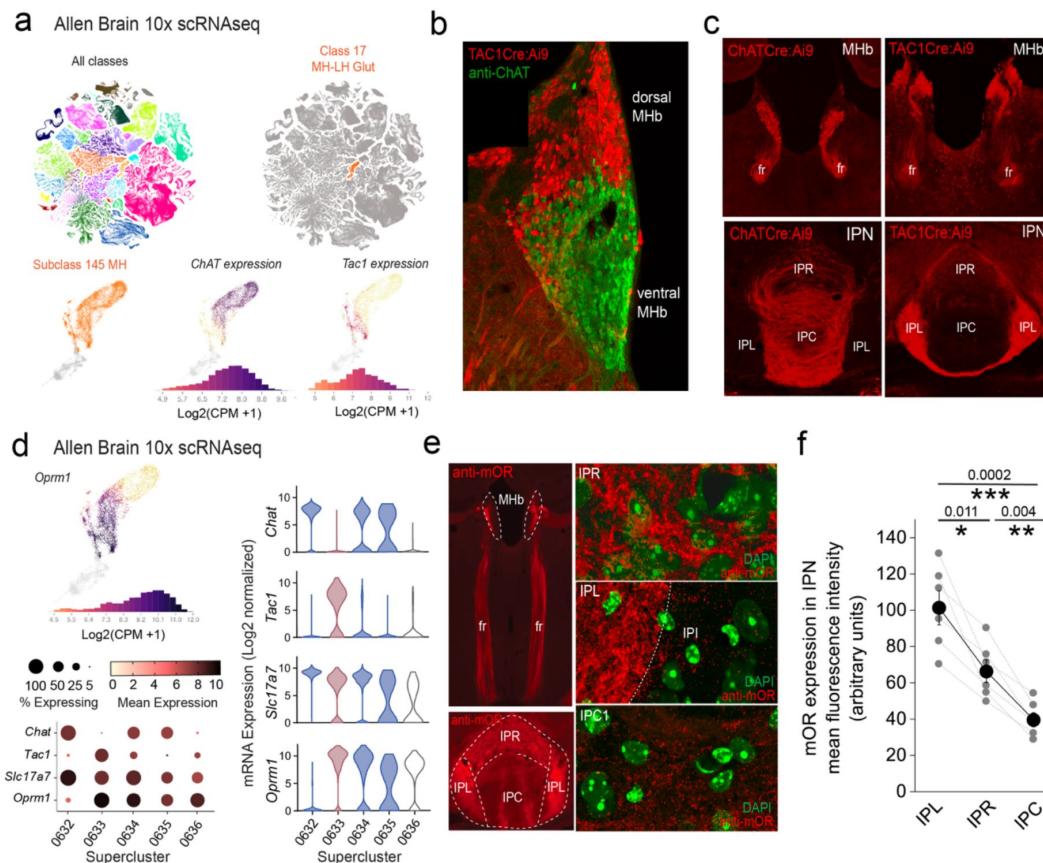


Fig. 1.

***OPRM1* gene and mOR protein expression in the habenula-interpeduncular axis.**

(a) 10X scRNAseq UMAP whole mouse brain showing all cellular classes (top left panel; 4.04 million cells) and Class 17 MH-LH Glut (right panel; corresponding to the mHb and lateral habenula; 10.8K cells). Bottom panels focus on individual mHb cells (Subclass 145 MH; 8K cells) indicating log2 mRNA expression of *ChAT* and *Tac1*. (b) Confocal image of a *Tac1Cre:Ai9* mouse immunostained with anti-ChAT illustrating the distinct distribution of substance P (red) and cholinergic neurons (green) in dorsal and ventral mHb, respectively. (c) Conditional td-Tomato expression in cholinergic (left panels; *ChATCre:Ai9*) and SP neurons (right panels; *Tac1Cre:Ai9*) illustrating their spatial location within the mHb with their axonal outputs in the fasciculus retroflexus (fr) and largely non-overlapping terminal axonal arborization patterns in IPN. (d) Profile of log2 mRNA expression of *OPRM1* in Subclass 145 MH (top left panel). Corresponding dot and violin plots depicting 3 CHAT (0632,0634,0635) and 1 TAC1 (0633) cluster with relative expression of *Slc17a7* (VGluT1) and *OPRM1* in each corresponding cluster. In the violin plots the CHAT clusters and TAC1 superclusters are color coded blue and red, respectively. (e) Endogenous mOR expression throughout the mHb and IPN axis (left panels; red) assessed by immunocytochemistry. High resolution airy scan images of mOR distribution in subdivisions of the IPN; rostral IPN (IPR), lateral IPN (IPC) and central IPN (IPL). Green fluorescence are DAPI stained nuclei. (f) Densitometry analyses of mOR expression in the various subfields of IPN. Data are from 2-4 slices containing IPN taken from each of 6 mice aged P40-P60. Data depicted in (a) and (d) are from the publicly available Allen Brain cell Atlas (<https://knowledge.brain-map.org/abcatlas>). See methods for further details.

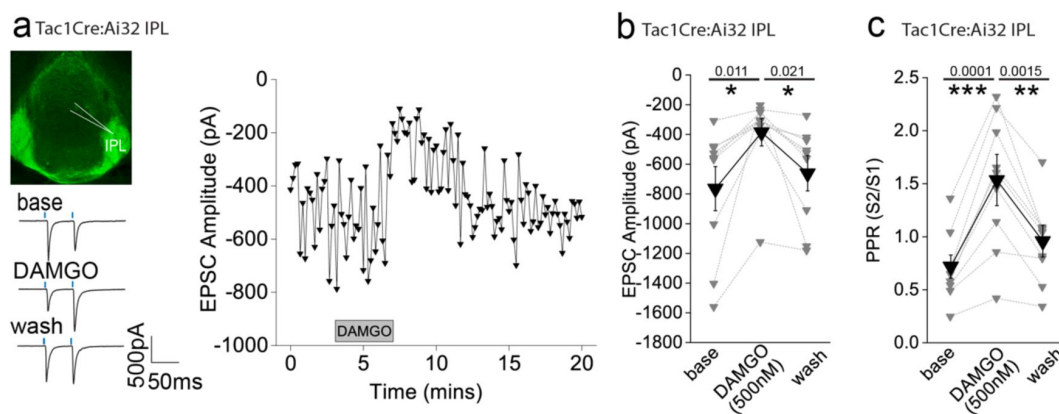


Fig. 2.

AMPA-mediated synaptic transmission in lateral IPN (IPL) mediated by substance P neurons is inhibited by MOR activation.

(a) Whole-cell voltage-clamp in adult (>p40) Tac1Cre:Ai32 mice (top left panel illustrating the axonal arborization of ChR2 expressing SP neuronal axons in IPN and the position of neuronal recording in IPL). Single voltage-clamp traces of light evoked AMPAR EPSCs (bottom left panel; 470nm light pulse; 2 stimulations at 20Hz, blue dashes)) and time course of peak amplitudes (right panel) under baseline, during application of 500 nM DAMGO and washout conditions. (b,c) Individual (grey filled symbols) and mean (black filled symbols) data of AMPAR EPSC amplitude and corresponding paired pulse ratios (PPR; n=9 recorded IPL neurons from 7 mice).

enhanced excitation-spike coupling in IPR GABAergic neurons

We next examined how mORs modulate mHb-IPN neurotransmission that occurs via the other major input mediated by cholinergic neurons. To this end, we employed ChATCre:ai32 mice and focused on the IPR subdivision (**Fig. 3a**) due to the relatively high levels of combined mOR expression and cholinergic neuronal innervation (**Fig. 1c,e,f**) in this subregion. Under our basal experimental conditions and in agreement with previous studies^{21,37,38} light evoked postsynaptic responses in the IPN mediated by mHb cholinergic neurons in response to either brief single or paired-pulse stimulation are solely mediated by AMPARs (see **Supplementary Fig. 2a** and **Supplementary Fig. 3a-c**). Remarkably, DAMGO application elicited a significant robust and reversible potentiation of light-evoked (470 nm; typically, 10-100 % arbitrary LED power equating to approximately 0.4 – 6.9 mW/mm²; CoolLED illumination system) AMPAR EPSC amplitude (**Fig. 3a,b**) in stark contrast to that seen with mHb TAC1-IPL glutamatergic signaling described previously (**Fig. 2a,b**). Notably, the accompanying PPR changes were not consistent with an increase in release probability raising uncertainty as to the synaptic locus of the mOR effect (**Fig. 2c**).

In these experiments we were “blind” to the molecular identity of the postsynaptic IPR neuron. Since a major population of neurons in this IPN subregion are of the somatostatin (SST) GABAergic subtype^{15,39}, it is likely that most of our recordings are from this subtype. Nevertheless, we also employed ChAT-ChR2 transgenic mice containing fluorescently reported SST neurons (i.e. ChATChR2:SSTCre:ai9; **Fig. 3d**) to allow targeted recordings specifically from these neurons. Our data clearly show that DAMGO robustly increases AMPAR EPSCs impinging on IPR RFP+ SST neurons (**Fig. 3d,e**) accompanied by a trending increase in PPR (**Fig. 3f**).

We also performed a series of additional experiments agnostic to the postsynaptic neuronal identity in IPC subdivision. Despite the relatively low expression levels of mOR in this subdivision of IPN (**Fig. 1e,f**) we observed a robust increase in AMPAR EPSC amplitude at a similar extent to that seen in IPR (**Supplementary Fig. 1**) thus extending this remarkable potentiation of glutamatergic synaptic transmission to all IPN subfields where cholinergic neuronal terminals reside.

We considered non-specific actions of DAMGO such as those that could be mediated by a possible direct modulation of ChR2 itself to explain this non-canonical observation. However, our previous data demonstrating the reduction of glutamatergic neurotransmission mediated by SP neurons by DAMGO described (**Fig. 2**) renders this possibility unlikely. Furthermore, a similar potentiation of cholinergic neuronal mediated glutamatergic transmission in the IPN upon activation of GABA_B receptors has demonstrated a propensity for this circuit to undergo such modulation in response to a Gi-linked receptor³²⁻³⁴. Nevertheless, despite this precedent, we employed a pharmacological approach to further validate this unexpected result. We show that lower concentrations of DAMGO (100 nM) and alternative mOR agonists, Met-enkephalin (Met-enk, 3 μM) and morphine (10 μM) all induce potentiation of ChAT neuronal glutamatergic transmission to similar extents (**Fig. 3g**). In addition, pre-treatment with the mOR selective antagonist (CTAP, 1 μM) completely prevents the DAMGO induced response (**Fig. 3g**). Together, this pharmacological battery of tests employing an experimental, an endogenous and clinical/recreational mOR agonists and selective mOR antagonism clearly implicate this opioid receptor in mediating a non-canonical potentiation of AMPAR EPSC amplitude elicited by mHb cholinergic neurons in IPN.

In several recordings from both “blind” patching of IPR neurons and in directly identified SST neurons, no measurable post synaptic AMPA receptor EPSC could be elicited following light-evoked stimulation of cholinergic axons even with the maximal possible light intensity available (i.e. 100 % arbitrary LED power; approximately 6.9 mW/mm²), and thus further interrogation was typically not performed. However, in a few instances, we nevertheless proceeded with DAMGO or Met-enk application and surprisingly, in a subset of these recordings (9/15 cells tested), a robust

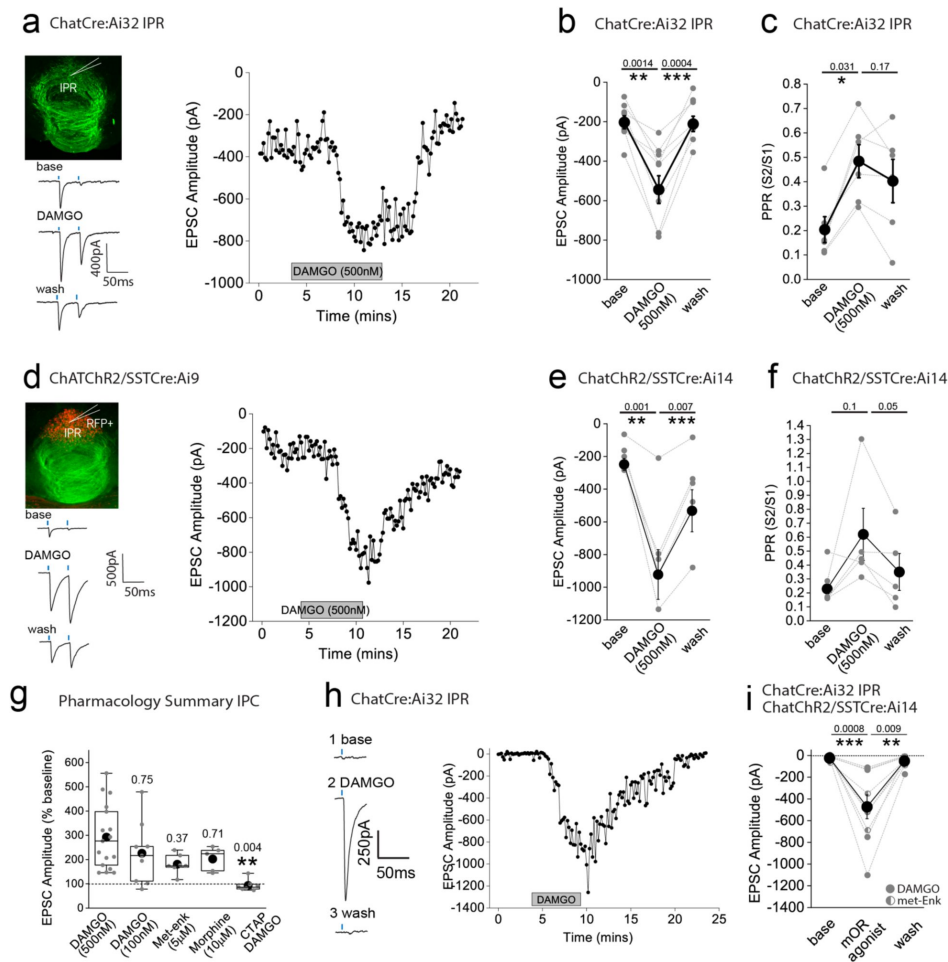


Fig. 3.

AMPA-mediated synaptic transmission in rostral IPN (IPR) mediated by mHB cholinergic neurons is potentiated by mOR activation.

(a) Whole-cell voltage-clamp in adult (>p40) ChatCre: Ai32 mice (top left panel illustrating the axonal arborization of ChR2 expressing cholinergic neuronal axons in IPN and the position of neuronal recording in IPR). Single voltage-clamp traces of light evoked AMPAR EPSCs (bottom left panel; 470nm light pulse; 2 stimulations at 20Hz, blue dashes) and time course of peak amplitudes (right panel) under baseline, during application of 500 nM DAMGO and washout conditions. (b,c) Individual (grey filled symbols) and mean (black filled symbols) data of AMPAR EPSC amplitude and corresponding paired pulse ratios (PPR; n=8 and 6 recorded IPR neurons from 7 and 5 mice for AMPAR EPSC amplitude and PPR, respectively). (d) Whole-cell voltage-clamp in adult (>p40) ChatChR2: SSTCre: Ai9 mice (top left panel illustrating the axonal arborization of ChR2 expressing cholinergic neuronal axons in IPN and the position of neuronal recording in RFP+ SST IPR neurons). Single voltage-clamp traces of light evoked AMPAR EPSCs (bottom left panel; 470nm light pulse; 2 stimulations at 20Hz, blue dashes) and time course of peak amplitudes (right panel) under baseline, during application of 500 nM DAMGO and washout conditions. (e,f) Individual (grey filled symbols) and mean (black filled symbols) data of AMPAR EPSC amplitude and corresponding paired pulse ratios (PPR; n=5 recorded IPR neurons from 4 mice). (g) Summary bar graph illustrating the effect of DAMGO (500nM and 100nM; n=16 recorded IPR neurons from 14 mice and n=9 recorded IPR neurons from 4 mice, respectively), met-enkephalin (5μM; n=7 recorded IPR neurons from 5 mice), morphine (10μM; n=5 recorded IPR neurons from 5 mice) and 500nM DAMGO in presence of mOR antagonist 1mM CTAP (n=7 recorded IPR neurons from 2 mice). (h) Voltage-clamp trace examples (single light stimulus; blue bar; left panel) and time course (right panel) under baseline, DAMGO and washout conditions in a recorded IPR neuron displaying no measurable baseline AMPAR EPSC in response to maximal light evoked stimulation (470nm; 6.9mW/mm²). (i) Individual (gray filled and half-filled symbols for 500nM DAMGO or 5μM Met-enkephalin respectively) and pooled (black filled symbols) data of light-evoked AMPAR EPSC amplitude during DAMGO application and washout (n = 6 and 3 recorded IPR cells from 6 and 3 mice for DAMGO and met-enkephalin application, respectively).

and reversible emergence of a significant AMPAR mediated EPSC was (**Fig. 2h,i**). Thus, these data clearly reveal that mOR activation invokes a synaptic mechanism that, at its most extreme, results in an “un-silencing” of ChAT neuronal glutamatergic transmission in IPN including in directly identified SST GABAergic neurons. Since, PPR measurements are only valid if the same population of release sites are assayed before and after experimental manipulation, the additional engagement of a putative reluctant vesicular pool³² by mOR activation could explain the discrepancy between the augmentation of EPSC amplitude and observed changes in PPR (**Fig. 3b,c,e,f**). However, future interrogation is necessary for definitive identification of the relative contributions of pre- and post-synaptic loci to the potentiation observed.

To date, we have examined the effect of mOR on synaptic transmission during paired pulse light activation delivered at 20 Hz. Although, mHb cholinergic neurons are capable of burst firing at much higher frequencies in response to afferent stimulation *in vivo*¹², they typically exhibit low frequency (i.e. <10Hz) intrinsic (i.e. independent of synaptic input) firing *in vitro*^{10,40–42}. In agreement with these previous studies, cholinergic neurons in ventral mHb (identified in ChATCre:Ai9 mice; **Fig. 1c**) spontaneously elicit action potentials measured in cell-attached recordings at an average of ~5 Hz (range 2 – 10Hz; **Fig. 4a**). Therefore, we assessed the role of mOR activation on glutamatergic transmission mediated by cholinergic neurons elicited by light stimulation trains delivered at this frequency. Interestingly, even with this relatively low frequency paradigm, glutamatergic transmission is extremely labile in nature as evidenced by the large and rapid depression of AMPAR EPSC during the stimulus train (**Fig. 4b,c**). Remarkably, activation of mORs essentially eliminates activity-dependent depression at this synapse (**Fig. 4b,c**). This switch in transmission dynamics greatly facilitates the probability of excitatory postsynaptic potential-spike coupling (ES coupling) in response to each stimulus within the train (**Fig. 4d-e**). Thus, these data clearly demonstrate that at physiologically relevant stimulus patterns, mORs serve to dramatically increase the fidelity of glutamatergic transmission in the IPR mediated by cholinergic neurons culminating in an enhanced recruitment of postsynaptic IPR neurons.

mORs constitute a molecular switch to alter the salience of glutamatergic transmission mediated by cholinergic and substance P neurons in the IPR

An intriguing inconsistency was observed regarding the DAMGO effect on light-evoked and spontaneous EPSCs (sEPSCs) impinging on IPR neurons. Under the experimental conditions employed sEPSC events were exclusively mediated by AMPARs as evidenced by their virtually complete cessation upon DNQX application (**Supplementary Fig. 2a**). Remarkably, in contrast to the previously described robust potentiation of light-evoked AMPAR EPSCs, activation of mORs significantly attenuates sEPSC frequency in agreement with a recent study³⁷, with no effect on amplitude (**Supplementary Fig. 2b-d**). Thus, in a single IPR postsynaptic neuron opposing effects on light-evoked and spontaneous AMPAR EPSCs is evident (**Supplementary Fig. 2e**). Although these data do not directly identify the origin and neuronal subtype(s) responsible for the sEPSCs measured, the incongruent effect of mOR activation on evoked versus spontaneous events led us to consider the possible existence of an additional afferent system impinging on IPR neurons. Prevailing circuit schemas based on axonal arborization patterns portray a mutually exclusive, non-overlapping afferent input to IPN mediated by cholinergic and SP neurons to IPR/IPC versus IPL, respectively. However, in the current study by employing high-resolution imaging in Tac1Cre:Ai32 mice a clear presence of synaptic bouton-like structures in the IPR is noted (**Fig. 5a,b**). These structures do not co-localize with endogenous ChAT, but do express the glutamate vesicular transporter, VGluT1 (**Fig. 5a,b**). This presence of putative presynaptic anatomical substrates points to the existence of a secondary input to IPR mediated by SP neurons distinct to the well-established one originating from mHb cholinergic neurons. Indeed, significant light-evoked AMPAR EPSCs in IPR neurons can be elicited in Tac1Cre:Ai32 mice with essentially

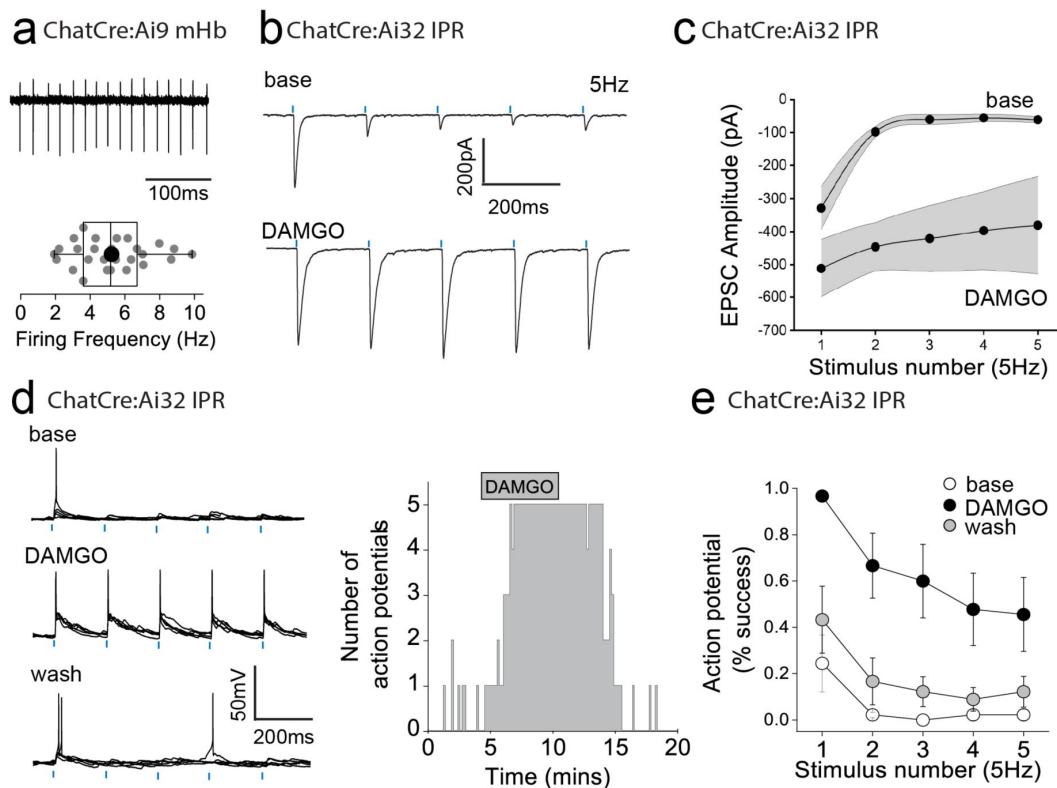


Fig. 4.

mOR activation increases fidelity of glutamatergic transmission mediated by mHb cholinergic neurons to augment excitation:spike coupling in postsynaptic IPR neurons.

(a) Single example trace showing spontaneous action potential firing in cell attached mode from a td-Tomato-positive ventral mHb neuron in the ChatCre: Ai9 mouse (top panel). Box plot and corresponding individual data of the spontaneous firing frequency of mHb cholinergic neurons (bottom panel; $n = 27$ recorded cells). (b) Single voltage-clamp traces of light-evoked (470nm, 5 pulses delivered at 5Hz; blue dashes) AMPAR EPSCs recorded from postsynaptic IPR neurons in adult ChatCre: Ai32 mice ($p > 40$) under baseline and following 500nM DAMGO application (left panel). (c) Pooled data (shaded area denotes SEM) of peak amplitude for each given stimulus in the 5Hz train during baseline and following 500nM DAMGO application ($n = 5$ recorded IPR neurons from 5 mice). (d) Single current-clamp traces example (5 consecutive overlaid sweeps) of light-evoked EPSP:action potential coupling (5 stimuli at 5 Hz; blue dashes) under baseline, 500nM DAMGO and washout conditions (left panel). Corresponding single example time course plot depicting number of light-driven EPSP-evoked action potential (right panel). (e) Mean data showing percentage success over 5 consecutive traces of EPSP:action potential coupling for each given stimulus in the 5Hz train under baseline (open symbols), 500nM DAMGO (black symbols) and washout (grey symbols) conditions ($n = 9$ recorded IPR neurons from 7 mice).

similar amplitudes across all LED powers employed to that seen in ChATCre:Ai32 mice (**Fig. 5c**). Thus, in contrast to recent ultrastructural EM analyses³², these data indicate a previously unidentified functional glutamatergic input to IPR mediated by SP neurons. Interestingly, DAMGO application significantly depresses the SP neuronal glutamatergic output in IPR and increases PPR (**Fig. 5d-f**) to a similar extent to that observed in IPL (**Fig. 2**; **Fig. 8i**). This is in direct contrast to the role of mORs in positively modulating the ChAT neuronal glutamatergic transmission in the IPR (**Fig. 3a,b,d,e**; **Fig. 8i**). Thus, for the first time we demonstrate both SP and cholinergic input to the same subregion of IPN and that mOR activation elicits diametrically opposing effect on transmission mediated by these respective presynaptic neuronal populations.

Dynamic regulation of the mOR elicited afferent specific switch in salience of the glutamatergic transmission to the IPR during adolescence

Our functional studies thus far have been restricted to sexually mature, adult mice (> p40). mOR expression emerges in many brain structures during prenatal development largely overlapping with their final adult distribution profile by mid/late gestation⁴³. In agreement, we reveal that mOR protein expression in IPR is present at appreciable levels from early postnatal stages (i.e. from P10 onwards; **Fig. 6a**). This begs the question whether modulation of interpeduncular synaptic transmission described during these critical early epochs mirrors that seen in the adult. To examine this, we extended our analyses to earlier time points (P15 to P40) and found that mOR modulation of glutamatergic transmission mediated by cholinergic neurons in the IPR undergoes a remarkable developmental regulation. This is characterized by an initial inhibitory effect on light-evoked AMPAR EPSCs transitioning to the previously described augmentation (**Fig. 6b,c,f**). In contrast, the glutamatergic input to the IPL and the newly discovered input to the IPR elicited by SP neurons is not developmentally regulated with synaptic transmission being consistently inhibited by DAMGO application at all ages tested (**Fig. 6d-f**). Thus, despite essentially similar levels of IPR mOR expression at the ages assayed (**Fig. 6a**), at earlier development stages opioids result in overall inhibition of afferent input to the IPR agnostic to afferent input with the emergence of the previously highlighted differential regulation of mHB cholinergic versus substance P neuronal output emerging during the late adolescent stages (**Fig. 6f**).

mOR induced potentiation of nicotinic receptor signaling in IPR is conditional on the removal of a molecular brake mediated by Kv1 channel function

In addition to mORs, this circuitry contains dense expression of nicotinic receptors located at pre- and postsynaptic sites^{10,44,45}. However, the ability to reliably evoke post-synaptic nicotinic receptor (nAChR) EPSCs in response to physiologically appropriate stimuli has been challenging. Indeed, many studies have resorted to high frequency and prolonged stimulation³⁶ or bath/puff applications of nicotine to investigate nAChR function in this circuit^{19,39,40,46-48}. Despite the faithful co-expression of endogenous ChAT in ChatCre:Ai32 expressing terminals located in the IPR (**Fig. 7a**), evoked transmission by brief light pulses (1-5ms) from cholinergic neurons predominantly elicits pure AMPAR EPSC in postsynaptic IPR neurons under our basal conditions in agreement with previous observations^{21,36,37,39} (**Supplementary Fig. 2a** and **Supplementary Fig. 3a-d**). Additionally, DAMGO, at concentrations that strongly potentiate glutamatergic transmission, does not result in a measurable postsynaptic nAChR EPSC after AMPARs are pharmacologically silenced (**Supplementary Fig. 3a-c**). Furthermore, repeated light stimulation at either 25 or 50Hz fails to elicit the emergence of any nAChR mediated EPSCs in the presence of DNQX (**Supplementary Fig. 3d,e**). Finally, the potentiated EPSC following mOR activation is purely AMPAR mediated as evidence by complete block by DNQX (**Supplementary Fig. 3f-h**). Together, our functional analyses demonstrate that brief pulses of light (1-5ms) in

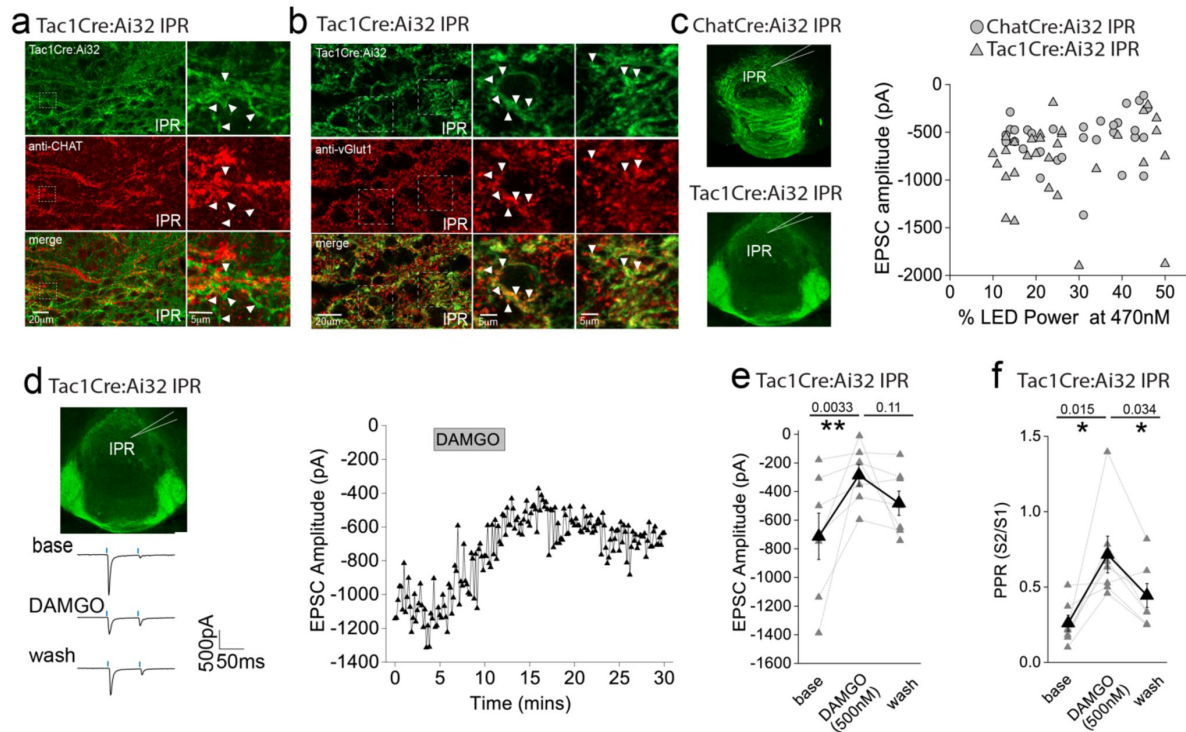


Fig. 5.

Effect of mOR activation on the novel functional SP neuronal mediated evoked AMPAR EPSCs in IPR.

(a) High resolution airyscan of the IPR in TAC1Cre:Ai32 mouse showing Chr2-expressing synaptic bouton like-structures (green) and endogenous ChAT expression (red). Right panels are magnified regions of the boxed area. Arrows indicate examples of non-overlap of TacCre:Ai32 boutons with ChAT. (b) High resolution airyscan of the IPR in TAC1Cre:Ai32 mouse showing Chr2-expressing synaptic bouton like-structures (green) and endogenous VGLUT1 expression via immunostaining (red). Right panels are magnified regions of the boxed area. Arrows indicate examples of expression of VGLUT1 within TacCre:Ai32 bouton structures. (c) Comparison of light-evoked AMPAR EPSC peak amplitude in postsynaptic IPR neurons mediated by either cholinergic (ChatCre:Ai32 mice; $n = 32$ recorded IPR neurons) and substance P (Tac1Cre:Ai32 mice; $n = 31$ recorded IPR neurons) mHB neurons at various arbitrary % LED (10–50% corresponding to an approximate power of 0.4 – 3.4 mW/mm²; right panel). (d) Whole-cell voltage-clamp in adult (>p40) TAC1Cre:Ai32 mice (top left panel illustrating the axonal arborization of Chr2 expressing SP neuronal axons in IPN and the position of neuronal recording in IPR). Single voltage-clamp traces of light evoked AMPAR EPSCs (bottom left panel; 470nm light pulse; 2 stimulations at 20Hz, blue dashes) and time course of peak amplitudes (right panel) under baseline, during application of 500 nM DAMGO and washout conditions (right panel). (e,f) Individual (grey filled symbols) and mean (black filled symbols) data of AMPAR EPSC amplitude and corresponding paired pulse ratios (PPR; $n=8$ recorded IPR neurons from 7 mice).

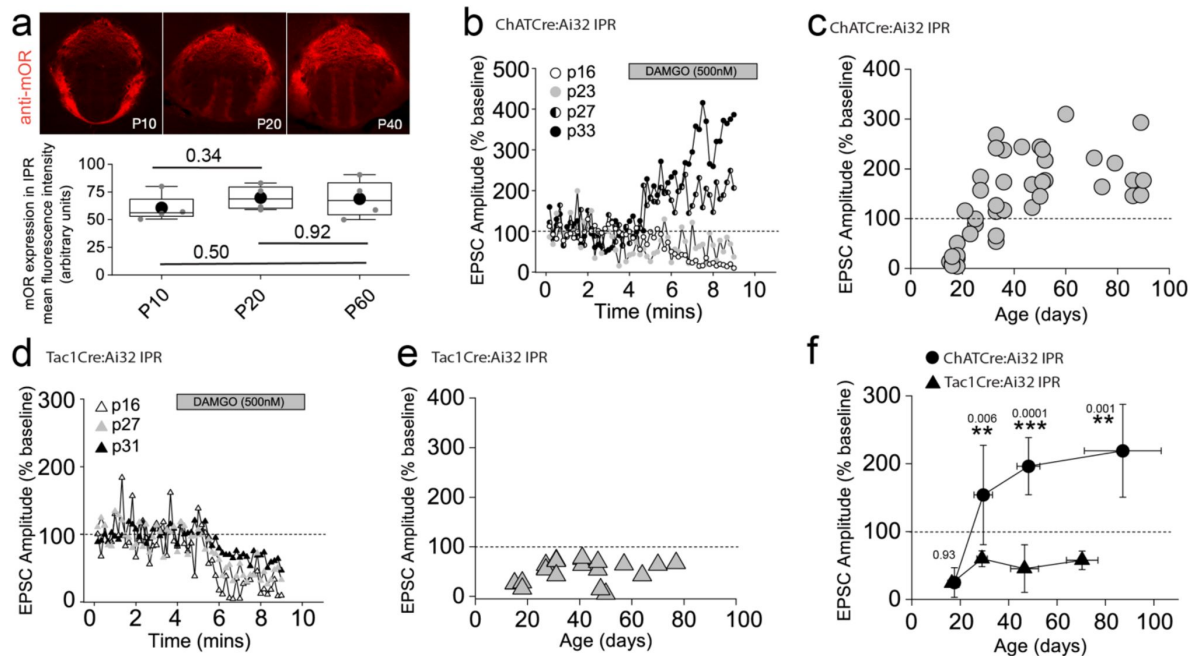


Fig. 6.

mORs constitute a developmentally regulated molecular switch altering the salience of neurotransmission in IPR mediated by substance P versus cholinergic neurons

(a) Confocal images of mOR protein expression in IPN during development (p10, p20 and p40). Densitometry analyses of mOR protein expression in IPR across development (measurements taken from 2 slices containing IPR from each of 2-4 mice for each age) (b) Single examples of the time course of light-evoked AMPAR EPSC peak amplitudes mediated by mHB cholinergic neurons following DAMGO application in postsynaptic IPR neurons at varying ages as indicated. (c) Individual data of the percent change of light-evoked cholinergic neuronal mediated AMPAR EPSC peak amplitude elicited following DAMGO application across all ages tested (n = 44 recorded IPR neurons). (d) Single examples of the time course of light-evoked AMPAR EPSC peak amplitudes mediated by mHB substance P neurons following DAMGO application in postsynaptic IPR neurons at varying ages as indicated. (e) Individual data of the percent change of light-evoked SP neuronal mediated EPSC peak amplitude elicited following DAMGO application across all ages tested. (f) Summary plot of the mean changes in the normalized AMPAR EPSC peak amplitude (% baseline) by DAMGO mediated by cholinergic and SP neurons binned at the following developmental epochs; p15-23 (postnatal), p24-34 (adolescent/pre-pubescent), p35-60 (adolescent/pubescent, sexual maturation) and >p60 (adult)⁹⁶. Total numbers of cells recorded = 39 and 21 cells from ChATCre: Ai32 and Tac1Cre: Ai32 mice, respectively. Error bars denote standard deviation of the mean. Note datapoints for ages >p40 in panels c and e are normalized data taken from recorded cells that were depicted in Figs. 3b and 5e as absolute peak amplitude changes, respectively.

ChatCre:ai32 mice does not result in sufficient ACh release, if any, to elicit measurable postsynaptic nAChR responses in response to high frequency stimulus regimens and even under conditions where mOR activation strongly enhances glutamatergic transmission.

Interestingly, cholinergic transmission at the neuromuscular junction (NMJ) can be boosted via potassium channel blockade with fampridine (4-aminopyridine) and has been clinically indicated in disorders such as myasthenia gravis and multiple sclerosis⁴⁹. Furthermore, low micromolar 4-AP in slices amplifies acetylcholine release, as assessed by Grab-ACh mediated fluorescence in the IPN^{50,51}. Experimentally, these concentrations are relatively selective in inhibiting K⁺-channels of the Kv1 family. Probing the publicly available 10X single cell RNAseq database provided by the Allen Brain Institute reveals that *KCNA2* (the gene encoding Kv1.2) is the most prevalently expressed in the designated ChAT Superclusters (**Fig. 1d**, **7b**; mean expression levels including zero values = 9.1, 6.8 and 4.1 in Superclusters 0632, 0634 and 0635, respectively). Furthermore, spatial interrogation via *in situ* hybridization (Allen Brain; <https://mouse.brain-map.org/gene/show/16263>) demonstrate the presence of appreciable *KCNA2* transcript levels in ventral habenula (**Fig. 7c**) a region that is enriched in cholinergic neurons (**Fig. 1b,c**). We therefore speculated that Kv1.2 block could result in enhancement of ACh release to perhaps reveal postsynaptic nAChR EPSCs in IPR neurons. Indeed, following block of glutamatergic transmission (DNQX + APV), subsequent application of 4-AP (50-100 μ M) or the more selective Kv1 channel antagonist, dendrotoxin- α (100 nM) results in the emergence of an EPSC elicited by brief single light pulses (**Fig. 7d**) and in response to trains of 5Hz stimulation (**Fig. 7e**). This emergent EPSC is mediated by nAChRs as evidenced by ~ 80-90% block by the nicotinic receptor antagonist, DH β E^{52,53} (2 μ M; **Fig. 7f**).

Interestingly, this experimental approach allows for the calculation of nAChR:AMPA peak amplitude ratio enabling an assessment of fast excitatory cholinergic transmission normalized to stimulus intensity required to produce a given AMPA EPSC response in an individual postsynaptic IPR neuron. Thus, the size of nAChR response can be directly compared across multiple recordings from differing slices, mice and experimental conditions. We first assessed the extent to which both 4-AP and DTX- α reveal light-evoked nAChR EPSCs finding no significant difference in the nAChR:AMPA ratio, demonstrating that the specific Kv1.1/Kv1.2 channel antagonist unmasks nAChR mediated synaptic transmission in a similar manner to low concentration 4-AP (**Fig 7g**). This, in combination with the predominant expression of *KCNA2* (**Fig. 7b,c**) and relative lack of other *KCNA* transcripts (**Fig. 7b**) in mHb ChAT neurons confirms our hypothesis that Kv1.2 plays a central role in unmasking nAChR-mediated synaptic transmission. In addition to ChatCre:Ai32 mice, we also employed the ChAT-ChR2 mouse. Not surprisingly, nAChR EPSCs could also be elicited in this mouse line following 4-AP application (**Fig. 7g**). However, the measured nAChR:AMPA ratio is significantly higher than that seen in ChatCre:ai32 mice (**Fig. 7g**) indicating a skew towards more prominent cholinergic transmission when compared to that mediated by glutamate. It is unclear as to the exact reason for this divergence, but it may be related to increased levels of the vesicular ACh transporter (vAChT) expression observed in ChAT:ChR2 mice⁵⁴. Although TAC1 mHb neurons also express *KCNA2* transcript albeit lower than in the CHAT neurons (data not shown), 4-AP application following block of glutamatergic transmission does not result in emergence of any additional EPSC in response to single stimuli in Tac1Cre:ai32 mice (**Fig. 7g**). Finally, application of a cholinesterase inhibitor (ambenonium, 50 nM) markedly prolongs the EPSC waveform with minimal effect on amplitude (**Fig. 7h**). Thus together, these results corroborate the role of Kv1.2 as a molecular brake, removal of which, results in the emergence of habenulo-interpeduncular synaptic transmission via postsynaptic nAChR signaling on IPR GABAergic neurons.

Having established the ability to reliably assess an otherwise reluctant cholinergic transmission, we next investigated how nAChR mediated signaling in the IPR is impacted by mOR activation. As with AMPAR mediated EPSCs, DAMGO produces a marked potentiation of light-evoked nAChR EPSC amplitude in IPR with the augmented EPSC being completely blocked by DH β E (**Fig. 8a,b**).

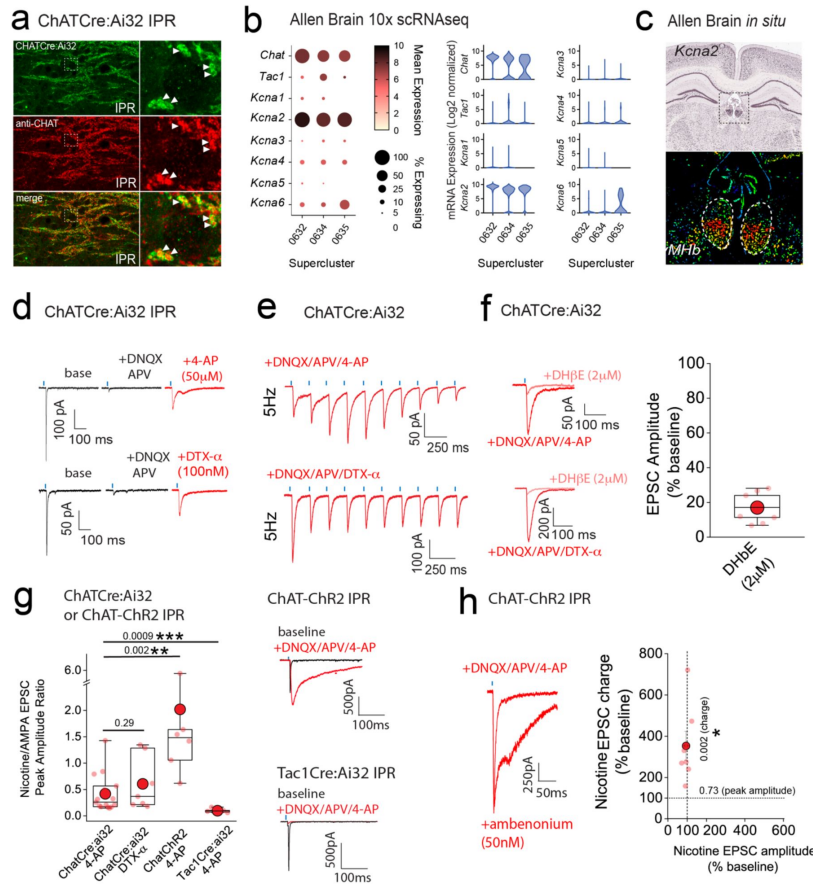


Fig. 7.

Kv1 channels constitute a molecular brake of nicotinic receptor mediated signaling in the IPN.

(a) High resolution airy scan images of IPR in ChATCre:Ai32 mice showing ChR2-expressing cholinergic boutons (green) and endogenous ChAT (red). Right panels are magnified regions of the boxed area in left panels. Arrows indicate faithful expression of ChAT within ChatCre:Ai32 boutons (cf. Fig. 5a). (b) Corresponding dot and violin plots illustrating the relative expression of *KCNA1-6* in CHAT superclusters only. (c) *In situ* hybridization for *KCNA2* mRNA (top panel) and corresponding pseudo-colored expression level (bottom panel) illustrating bias towards ventral mHb. Data are from the Allen Brain Institute (<https://mouse.brain-map.org/gene/show/16263>). (d) Voltage-clamp example traces of light-evoked EPSCs mediated by cholinergic mHb neurons (left panel) under baseline, following 10μM DNQX/100μM DL-APV plus 50μM 4-AP (top panel) or plus 100nM dendrotoxin-α (bottom panel). (e) Voltage-clamp example traces of 5Hz trains of light-evoked EPSCs (10 stimuli) mediated by cholinergic mHb neurons in the presence of 10μM DNQX/100μM DL-APV and 100μM 4-AP (top panel) or DTX-α (bottom panel) in a ChATCre:Ai32 mouse. (f) Single light stimulus evoked EPSC in the presence of 10μM DNQX/100μM DL-APV and 100μM 4-AP (top left panel) or DTX-α (bottom left panel) in the absence or presence of 2μM DHβE. Box plot with individual data of the percentage inhibition of the light-driven EPSC peak amplitude mediated by cholinergic neurons in the presence of 10μM DNQX/100μM DL-APV/50μM 4-AP or DTX-α (n = 9 recorded IPR neurons from 9 mice). (g) Box plot of the nicotine/AMPA peak amplitude ratio within individual recorded IPR neurons percentage as measured under baseline (AMPA EPSC) and in the presence of 10μM DNQX/50μM DL-APV/100μM 4-AP (nAChR EPSC; n = 15 recorded IPR neurons from 15 ChATCre:Ai32 mice) or 100nM DTX-α (nAChR EPSC; n = 7 recorded IPR neurons from 5 ChATCre:Ai32 mice). nAChR/AMPA peak EPSC amplitude ratios were also performed in ChAT-ChR2 (n = 6 recorded IPR neurons from 6 mice) and Tac1Cre:Ai32 mice (n = 6 recorded IPR neurons from 6 mice). Voltage-clamp example trace of light-evoked EPSCs under baseline and after addition of 10μM DNQX/50μM DL-APV/100μM 4-AP in ChAT-ChR2 (top right panel) and Tac1Cre:Ai32 (bottom right panel). (i) Voltage-clamp example trace of light-evoked nAChR-mediated EPSCs mediated by cholinergic mHb neurons under baseline and following application of 50nM ambenonium (left panel) in a ChAT-ChR2 mouse. Scatter plot of the individual (light red symbols) and pooled (red symbol) percentage change in nAChR EPSC peak amplitude versus EPSC charge (measured over the first 500ms duration of the EPSC) in each individual recording (n = 7 recorded IPR neurons in 5 mice; right panel). Data in (b) and (c) are from the publicly available Allen Brain Cell Atlas (<https://knowledge.brain-map.org/abcatlas>) and the Allen Brain Map (Allen Brain; <https://mouse.brain-map.org/gene/show/16263>), respectively. See methods for further details.

The mOR-mediated potentiation of both the glutamatergic and cholinergic output is essentially similar (**Fig. 8i**). Till now, we have utilized optogenetic approaches to assay synaptic transmission in the IPN. Interestingly, following acute block of Kv1, nAChR EPSCs could also be elicited via electrical stimulation (**Fig. 8c**). Further, DAMGO reliably potentiated both electrical and light evoked nAChR EPSCs in a single IPR neuron to similar extents, a response that is effectively blocked by DHβE (**Fig. 8c-e**). Finally, we tested the molecular specificity of the DAMGO response employing ChAT-ChR2:global *OPRM1* knockout mice. In these mice DAMGO application is ineffective in augmenting glutamatergic or cholinergic mediated neurotransmission in IPR whilst the potentiation upon GABA_B receptor activation previously described^{32,34,55} remains intact (**Fig. 8f-h**; note that in these experiments CGP55845A which is routinely included in all other experiments was omitted). This genetic approach complements our previous pharmacological data (**Fig. 3g**) to directly implicate mORs in the potentiation of mHb-IPN transmission. The overall effects of mOR activation on synaptic transmission mediated by the distinct mHb afferent systems in the various subdivisions of IPN tested is summarized in **Fig. 8i**.

Discussion

In the current study, we reveal for the first time a remarkable augmentation of glutamatergic/cholinergic co-transmission by mHb cholinergic neurons following mOR activation in the IPN. However, the underlying cellular/network mechanisms responsible remain unclear. The most parsimonious explanation is that mORs in proximity to the release machinery of cholinergic nerve terminals²³ mediate this effect. However, one must consolidate the fact that a Gi-linked receptor, which typically serve to directly inhibit release, results in a seemingly paradoxical potentiation. Functional and mechanistic studies have elegantly demonstrated that GABA_B receptors (another member of the Gi-linked subfamily) trigger acute molecular and structural adaptations resulting in enhanced Ca²⁺ influx and a switch of transmission modes within single presynaptic terminals to ultimately increase neurotransmitter release^{32,33}. Although, these observations set a precedent for potentiation by Gi-linked receptor activation in the IPN similar experimental approaches to that employed in the examination of the GABA_B-receptor mediated potentiation^{32,33,51} are required to definitively implicate overlapping mechanisms. In other brain regions, a canonical inhibitory influence of mORs elicits overall network excitation via disinhibition^{56–59}. Furthermore, numerous studies have revealed an intricate interplay involving diverse neuromodulatory components intrinsic to the IPN that serve to regulate synaptic transmission^{15,55,60}. Thus, taking the results of the current study in isolation, we cannot discount the possibility that the potentiation observed here may result via the activation of mORs residing on other neural elements within the IPN microcircuitry such as postsynaptic neurons or glia, for instance. However, it must be noted that a recent study employing a viral strategy to elicit conditional knockout of mORs in *OPRM1*-expressing mHb neurons specifically, prevents the DAMGO mediated potentiation of glutamatergic transmission onto the IPN indicating a presynaptic locus³⁷. Regardless, of the exact underlying cellular and/or network mechanisms our data extend the previously described Gi-linked receptor potentiation of neurotransmission in the IPN^{32,33,51} to mORs, a predominant target of a societally relevant and prevalently misused class of drugs. It would be of interest to determine whether GABA_B receptor activation exerts a similar inhibitory influence as mORs on the newly discovered SP neuronal mediated transmission to the IPR. Similarly, does the modulation of ChAT neuronal glutamatergic output undergo a developmentally regulation in a similar manner to that observed with mORs? These additional functional comparisons of the synaptic influences of these two distinct Gi-linked receptors may shed light as to the similarity, or lack thereof, regarding the underlying cellular mechanisms.

Recent generation of transgenic mice^{24,61,62} have permitted manipulation of mOR-expressing neurons and mOR receptors including those expressed specifically within the mHb/IPN. Emerging behavioral studies adopting such conditional genetic approaches have

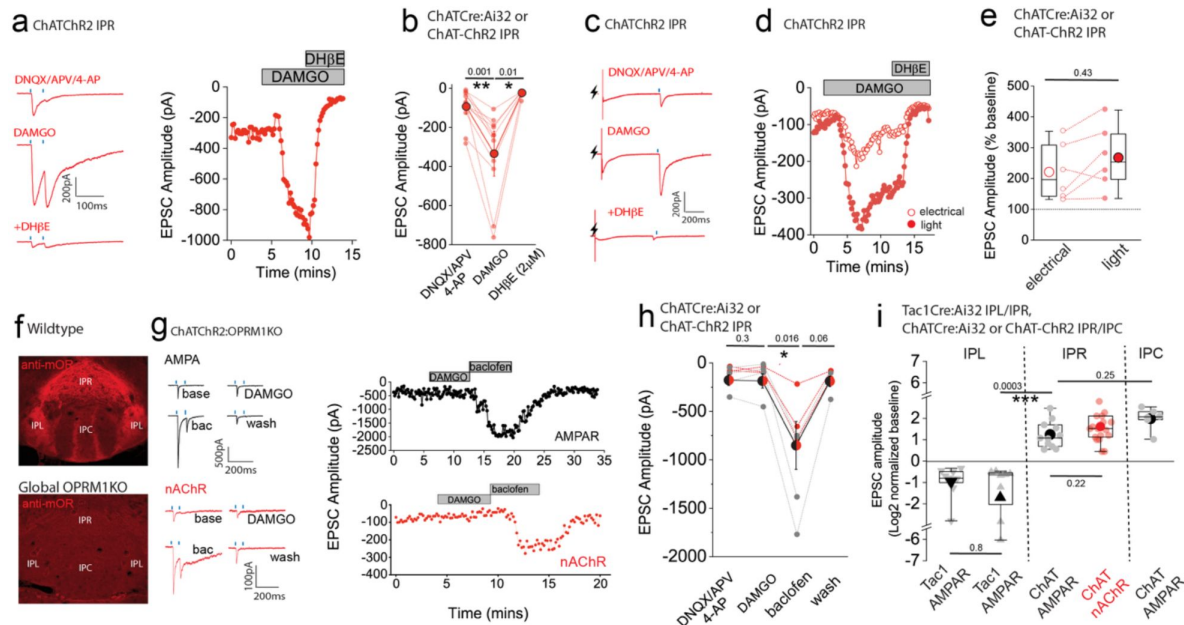


Fig. 8.

mOR potentiates nAChR EPSC amplitude revealing an interplay between opioid and cholinergic systems in the habenulo-interpeduncular axis.

(a) Voltage-clamp example traces of light-evoked nAChR EPSCs (2 stimuli at 20Hz; blue dashes) mediated by cholinergic mHb neurons (left panel) and time course of peak amplitude (right panel) under baseline, 500nM DAMGO and 500nM DAMGO plus 2μM DHβE conditions. (b) Individual and mean (red filled symbols) data of nAChR EPSC peak amplitude. (c,d) Voltage-clamp example traces of simultaneous electrical (lightning symbol) and light-evoked (blue dash) nAChR EPSCs and time course of peak amplitude (open and filled symbols representing electrical and light-evoked peak amplitude of nAChR EPSCs, respectively) under baseline, 500nM DAMGO and 500nM DAMGO plus 2μM DHβE conditions. (e) Percentage change in electrical and light-evoked nAChR EPSC peak amplitude elicited by 500nM DAMGO in each individual recorded IPR neuron (n = 6 recorded IPR neurons from 6 mice). (f) Confocal images of endogenous mOR protein expression in IPN of WT and homozygote OPRM1 KO mice. (g) Voltage-clamp example traces of light-evoked AMPA and nAChR EPSCs (2 stimuli at 20Hz; blue dashes) mediated by cholinergic mHb neurons (top and bottom left panels, respectively) and time course of peak amplitude (right panels) under baseline, 500nM DAMGO, 1μM baclofen and washout conditions in ChAT-ChR2:OPRM1KO mice. (h) Individual and mean data of light-evoked AMPAR (black symbols; n=4 from 4 mice) and nAChR (red symbols; n=3 from 3 mice) peak amplitude in response to DAMGO and baclofen application in ChAT-ChR2:OPRM1KO mice. Note in 2 of the 4 AMPAR EPSC recordings washout of the baclofen effect was not performed. (i) Summary box plot with individual data of the percentage normalized changes (log2) of AMPAR and nAChR peak amplitudes in response to mOR activation mediated by SP and cholinergic neurons in various subdivisions of IPN tested. Note that the normalized data in this panel are replotted from the absolute peak EPSC amplitude changes mediated by mOR agonists in [Figs. 2b](#), [3b](#), [3f](#), [5e](#), [7b](#) and [Supplementary Fig. 1a](#).

highlighted an important role of mORs within this circuitry in mediating varying aspects of reward and aversion^{25,26}. Unsurprisingly, it has been concluded that the observed behavioral effects of such manipulations are due to perturbations of a generalized inhibition mediated by mORs within the mHb/IPN circuit. Here, we reveal a functional dichotomy characterized by an inhibitory yet excitatory influence on neurotransmission mediated by SP and cholinergic neurons, respectively. These distinct neuronal populations participate in and drive specific behavioral facets demonstrating a division of labor. For example, SP neuronal activity positively correlates with reward outcome, history and hedonic value^{63,64}, whereas that of cholinergic neurons is reduced during behaviors associated with reward⁶³. mHb cholinergic neurons have also been linked to negative affect including anxiety and depression^{42,65,66} that promote drug-seeking behaviors during withdrawal⁶⁷. We reveal for the first time that the IPR subdivision receives functional glutamatergic afferent input from SP neurons that based on previous anatomical interrogation were considered to exclusively target the IPL⁶⁸. Interestingly, the novel opposing effect on transmission extends to the IPR thus demonstrating a role for mORs in altering the relative salience of these distinct afferent systems with regard recruitment of common downstream IPR neurons. The IPR houses somatostatin (SST) GABAergic neurons whose afferents impinge on the raphe nucleus and lateral dorsal tegmental nucleus, the latter influencing the VTA and, in turn, its downstream structures^{15,39,69}. Thus, careful consideration regarding future examination of behavioral outcomes resulting from the described contrasting effects of mOR activation on the parallel processing of reward (mHB SP) and anti-reward (mHB ChAT) in this circuit is essential. One limitation of the current study arises from the sole utilization of a transgenic approach to selectively assay light-evoked synaptic transmission from ChAT and SP neuronal populations, respectively. Although the mHb provides most of the afferent input to the IPN, this approach does not exclude possible activation of additional inputs from other brain regions^{22,70,71}. This caveat can be circumvented by use of viral strategies to express ChR2 solely in the mHB^{21,37,39}. However, it is unclear if neuronal inputs from these possible alternate sources^{22,70,71} are glutamatergic in nature and mediated by a TAC1/OPRM1-expressing neuronal population. Nevertheless, to definitively identify the described novel input as one that originates from mHb SP neurons will require future use of such viral strategies.

Throughout this study we contextualize the effect of mOR activation in the IPN primarily through the lens of exogenously introduced opioids. However, physiological activation of mORs can be mediated to differing extents by endogenous ligands such as dynorphin or enkephalin. Indeed, we demonstrate that met-enkephalin effectively boosts co-transmission from cholinergic neurons in a similar manner to morphine. Interestingly the IPN is home to a population of pro-enkephalin (PENK)-expressing neurons and detectable met-enkephalin immunoreactivity^{23,72}. In other brain regions, manipulation of PENK neuronal activity and or ablation of PENK itself elicits mOR mediated circuit modulation and behavioral alterations^{73–75}. It is anticipated that activation of mORs by local endogenous opioid release yields similar complex effects to those following exogenous application of mOR agonists described here. Thus, this physiological route will have major implications concerning the role of mHB/IPN mORs in mediating the positive reinforcing effects of non-opioid drugs of abuse (e.g. nicotine, alcohol, amphetamines) and other “natural” rewarding stimuli (e.g. such as those associated with social interaction, exercise and food intake) thus extending the relevance of our study to these additional modalities.

Another novel finding in the current study relates to the identification of a “molecular brake” that exerts strong control over nAChR signaling impinging on postsynaptic IPN neurons. Specifically, compromising the function of delayed rectifying K⁺-channels containing Kv1.2 subunits results in emergence of nAChR EPSCs in response to light and electrical stimuli delivered with physiological paradigms likely thru increased axonal/bouton excitability and calcium influx via action potential waveform modulation. Kv1.2 function can be bidirectionally impacted by use-dependent mechanisms or through secondary activity mediated cascades resulting in post-translational

modifications (e.g. phosphorylation state) that impact its function and/or trafficking^{76–78}. These plausible endogenous mechanistic routes could serve to titer the strength nAChR mediated transmission in the IPR.

The resident nAChRs in the mHb/IPN are primarily encoded by the CHRNA3/B4/A5 gene cluster and the role of this circuitry in nicotine use is well characterized. For example, increased propensity for nicotine abuse in humans is associated with dysregulation of nAChR function precipitated by single nucleotide polymorphisms of this gene cluster^{79–81}. Experimental manipulation of nAChR signaling in the mHb/IPN generates phenotypes associated with various aspects of nicotine consumption in mouse models^{53,82–85}. Furthermore, chronic nicotine results in adaptations in nAChR expression and function that can exacerbate continued nicotine consumption^{40,48,86,87}. Together, these studies establish a link between nAChR mediated signaling within this circuitry and prevalence of nicotine misuse. Here we highlight an intriguing interplay between opioid and cholinergic systems within the mHb/IPN axes. Our data clearly demonstrate that mORs and Kv1.2 together comprise synergistic molecular targets that, in addition to others previously identified³⁴, when leveraged in tandem can manipulate nAChR mediated signaling to yield potential interventions for nicotine overuse.

Adolescence is a critical period for many aspects of brain and social development and represents a high-risk demographic group for drug use developing into long-term addiction⁸⁸. Propelled by the societal introduction of highly potent synthetic morphine analogs (e.g. fentanyl), one of the many devastating sequelae of the well-documented opioid crisis is an alarming high rate of overdose deaths not only in adults but also during vulnerable teenage years. Of note is the striking valence conversion from depression to potentiation of mHb ChAT neuronal signaling in the IPR occurring around the late postnatal stages. This is in stark contrast to that seen with the newly described SP neuronal mediated transmission in this same IPN subdivision where mOR activation is inhibitory at all life stages assayed. Thus, during postnatal development opioids acting via mORs result in a blanket inhibition of these two distinct inputs prior to the establishment of the afferent specific modulation in the adult. Interestingly, numerous studies have highlighted changes in the role of mORs in reward associated behaviors coinciding with similar periods^{89,90}. An attractive hypothesis is that coordinated modifications at the cellular, molecular and/or network level underlie this dynamic developmental regulation. Thus, our circuit analyses provide insights into the neural correlates for the differing propensity of mORs to regulate not only positive reinforcement but also the negative effects associated with withdrawal at various stages of development⁹¹. Whether these adaptations underlie the increased propensity for substance abuse during development remains to be ascertained. Furthermore, opioid exposure during early developmental epochs imparts long lasting effects on the brains reward circuitry³. Thus together, it is imperative to consider the newly described developmentally regulated impact of mORs in the modulation of the mHb/IPN circuitry. This is of particularly relevance with respect to generation of potential preventative treatments for the deleterious consequences of fetal opioid exposure and SUDs in high-risk juveniles.

In addition to a central role in addiction, the mHb/IPN circuitry encodes fear associated behaviors^{32,33,92–94}. Interestingly, prevention of GABA_A-receptor function in mHb cholinergic neurons or increasing IPN neuronal activity facilitates and augments expression of fear extinction, respectively^{32,33}. Dysregulation of this behavioral aspect is a key underlying cause of post-traumatic stress disorder (PTSD)⁹⁵. Based on this previous work, the novel mOR mediated potentiation of the mHb cholinergic transmission and hence IPN neuronal recruitment described here constitutes a parallel, yet alternative circuit mechanism involved in the extinction of learnt fear. Thus, pharmacogenetic manipulation of mOR mediated signaling specifically in the IPN may constitute a viable approach to alleviate conditions associated with abnormal fear processing such as those seen in PTSD.

In summary, we highlight several previously undescribed and hence unappreciated roles of mORs in the dynamic regulation of synaptic transmission impinging on IPN GABAergic neurons following exposure to exogenous opioids. Although additional studies are needed to reveal the underlying cellular mechanisms responsible for these complex, divergent facets of opioid mediated effects in the habenulo-interpeduncular axis, the current detailed functional analyses lays out a valuable roadmap. One that necessitates consideration in future interrogation concerning the role of these receptors in this relatively understudied brain circuitry that encodes aspects of emotion, hedonic and addiction behaviors.

Methods

Animals

All experiments were conducted under an active animal protocol approved by the Animal Care Use Committee of the National Institute of Child Health and Human Development. All transgenic mice were attained from The Jackson Laboratory (ME, USA) and were as follows: ChATCre (strain #031661), Tac1Cre (strain #021877), SSTCre (strain #018973); Ai9 (strain #007909), Ai32 (strain #024109), ChAT-ChR2-EYFP (strain #014546); OPRM1 KO (strain #007599). For conditional expression of td-Tomato (Ai9) or ChR2 (Ai32) both male and female breeders were homozygous, and all immunocytochemical or electrophysiological experiments were performed in their progeny that were heterozygous for both the Cre and floxed alleles. Heterozygous ChAT-ChR2 mice were used throughout the study either alone or crossed with the OPRM1KO mouse (homozygous) or SSTCre:ai9 (heterozygous for both alleles). Both male and female mice were investigated, and the data were pooled.

Reagents

Pharmacological reagents used in this study are as follows: DNQX (Cat. No. 2312/10), DL-APV (Cat. No. 0105/10), CGP55845A (Cat. No. 1248/10), Picrotoxin (Cat. No. 1128), bicuculline (Cat. No. 0109/10), DAMGO (Cat. No. 1171/1), CTAP (Cat. No. 1560/1) and DH β E (Cat. No. 2349) were all purchased from Biotechne/Tocris (MN, USA); Dendrotoxin- α (Cat. No. D-350) was purchased from Alomone Labs (Jerusalem, Israel). met-enkephalin (Cat. No. M6638) was purchased from MilliporeSigma (MA, USA). Ambenonium (Cat. No. sc-203507) was purchased from Santa Cruz Biotechnology (TX, USA) Morphine Sulphate (Cat. No. NDC: 0641-6125) was from Hikma (NJ, USA) and attained via the NIH Division of Veterinary Resources (DVR) Pharmacy.

Primary and corresponding secondary antibodies used in this study for immunocytochemistry and the working dilutions employed are as follows: Guinea pig anti-RFP (Synaptic Systems, Cat. No. 390005, 1:500) with CF555-conjugated Donkey anti-Guinea Pig IgG (Biotium Cat. No. 20276, 1:1000). Chicken anti-GFP (AvesLabs, Cat. No. GFP-1010, 1:1000) with CF488-conjugated Donkey anti-Chicken IgY (Biotium Cat. No. 20166, 1:1000 working dilution). Rabbit anti-MOR (ABCAM, Cat. No. ab134054, 1:500) with CF555-conjugated Donkey anti-Rabbit IgG (Biotium cat#20038, 1:1000). Goat anti-ChAT, Millipore (Cat. No. AB144P, 1:1000) with CF555-conjugated Donkey anti-Goat IgG (Biotium, Cat. No. 20039, 1:1000 working dilution).

Immunocytochemistry and Imaging

Mice ($P > 15$) were perfused trans-cardinally using 4% PFA and dissected brain tissues were post-fixed in 4% PFA for 24 hours at 4 °C. For P10 mice, brains were removed and drop fixed in 4% PFA for 24 hours at 4 °C. Fixed brain tissues were thoroughly washed in 1x phosphate buffer (PB) followed by cryopreservation using 30% sucrose. 50 μ m coronal sections were made on a frozen microtome. To perform floating section IHC, brain slices were washed with 1x PB at room temperature for 1 hour with 2-3 changes of 1x PB followed by blocking and permeabilizing in Blocking Solution (1x PB + 10% donkey serum + 0.5% Triton X-100) at room temperature for at

least 2 hours. Blocked brain slices were incubated in primary antibodies, which were diluted to working concentration using Antibody Solution (1x PB + 1% donkey serum + 0.1% Triton X-100), at 4 °C for 24-48 hours. After wash with 1x PB at room temperature for 15 minutes with 3 repeats, brain slices were incubated in secondary antibodies diluted with Antibody Solution at room temperature for 1 hour, followed by DAPI staining at 1µg/ml for additional 15 minutes. After washed with 1x PB at room temperature for 15 minutes with 3 repeats, brain slices were mounted on gelatin coated slides followed by air drying, covered with No.1.5 cover slides ProLong Glass Antifade Mountant (ThermoFisher Scientific, Cat#P36984), cured in darkness overnight before imaging. Confocal images were acquired on a Zeiss LSM900 using ZenBlue software. Light path and scan settings were conFig.d for best signal with multiplex channels. 10x and 20x confocal images of whole habenula and IPN were acquired as tiles with z-steps of 3µmx5 and 0.9µmx20, respectively. Airy scan images were acquired at 63x using multiplex 2Y mode to achieve super resolution. 2µm depth of images were captured at z-steps of 0.2µmx10, with or without tiling. All comparable images of tissues with different genotypes or developmental stages were acquired using the same light path configuration, channel input, gain and offset. Images were imported into Fiji and Adobe Photoshop for processing and densitometry measurements where applicable.

Electrophysiology

P15-P91 mice (for details refer to the results section) were anesthetized with isoflurane, and the brain removed in ice-cold partial sucrose substituted ASCF (ssACSF) solution containing (in mM): 90 Sucrose, 80 NaCl, 25 NaHCO₃, 1.25 NaH₂PO₄, 3.5 KCl, 4.5 MgCl₂, 0.5 CaCl₂, 10 glucose, saturated with 95% O₂ and 5% CO₂ (pH 7.4; osmolarity 300-310 mOsm). Coronal sections containing wither the mHb or IPN (270 – 300 µm) were cut using a VT-1000S vibratome (Leica Microsystems, Bannockburn, IL) in ice-cold ssACSF. The slices were allowed to recover in the ssACSF at 31-33°C for 20-30 minutes followed by an additional 20 minutes at room temperature. Slices were then transferred to our standard extracellular solution (ACSF) of the following composition (in mM): 130 NaCl, 24 NaHCO₃, 3.5 KCl, 1.25 NaH₂PO₄, 2.5 CaCl₂, 1.5 MgCl₂, and 10 glucose, saturated with 95% O₂ and 5% CO₂ (pH 7.4; osmolarity 300-310 mOsm) for storage until electrophysiological recording. Slices were transferred to an upright Olympus BX51WI microscope and visualized with infrared differential interference contrast microscopy and perfused (2-4 ml/min) with the above extracellular solution. Unless otherwise indicated the recording extracellular solution was routinely supplemented with 2µM CGP55845A, 50µM picrotoxin and 10 µM bicuculline (Tocris Science or Ascent Scientific).

All recordings were performed at 31–33°C with electrodes (3–5 MΩ) pulled from borosilicate glass (World Precision Instruments, Sarasota, FL) filled with one of two intracellular solutions (in mM): (i) 135 CsMeSO₄, 8 KCl, 4 MgATP, 0.3 NaGTP, 5 QX-314, 0.5 EGTA and 3 mg/ml biocytin. (ii) 150 K-gluconate, 3 MgCl₂, 0.5 EGTA, 2 MgATP, 0.3 Na₂GTP, 10 HEPES and 3 mg/ml biocytin. pH was adjusted to 7.3 with KOH or CsOH and osmolarity adjusted to 285–295 mOsm. Whole-cell patch-clamp recordings were made using a Multiclamp 700B amplifier (Molecular Devices, Sunnyvale, CA). Signals were filtered at 4 – 10 kHz and digitized at 10 – 20 kHz (Digidata 1322A and pClamp 10 Software; Molecular Devices, Sunnyvale, CA). For cell attached recordings pipettes were filled with our standard extracellular solution.

For optogenetic stimulation of Chr2 light stimuli at a wavelength of 470nm and duration of 1-5 ms were delivered through the 40X water immersion objective using a CoolLED pE-4000 Illumination system (Andover, UK). Arbitrary LED power typically ranged between 10 – 100% corresponding to ~ 0.4 – 6.9 mW/mm² as measured with a digital power meter (Thor Labs, NJ, USA). sEPSC analyses was performed on post hoc filtered episodic traces (Clampfit; Bessel 8-pole, –3db cutoff 1000Hz). When applicable, electrical stimulation (0.1 ms; typically, 100-700 µA) were performed by placement of a tungsten bipolar electrode in the IPR (World Precision Instruments FL, USA; Catalog # TST33A05KT) and a constant current stimulus isolator (World Precision Instruments; Catalog # A385 or A365). Frequency of sEPSCs and average sEPSC amplitude were determined over 5 consecutive sweeps totalling a period of 4 seconds (i.e. 0.8s per sweep). sEPSC were detected via a

thresholding procedure set at 4 times the RMS baseline noise. Note that only neurons with a baseline sEPSC frequency of > 8Hz were selected for further interrogation. Analyses of electrophysiological data was performed in Clampfit (Molecular Devices, Sunnyvale, CA) and curated in Microsoft Excel spreadsheets.

Analyses of publicly available single-cell

RNA sequencing transcriptomic data

We utilized the Allen Institute dataset of the single-cell transcriptomes of ~4 million cells across the entire mouse brain³⁵. To probe the region and genes of interest, we adapted Python notebooks provided for accessing the Allen Brain Cell (ABC) Atlas (https://alleninstitute.github.io/abc_atlas_access/intro.html). The log2-normalized expression matrices were downloaded and filtered to reach the cell types of interest within the habenula class “17 MH-LH Glut” (~10.8k cells). For the current study we focused our analyses on the medial habenula subclass “145 MH Tac2 Glut” (~8k cells). Within this subclass, we probed the 5 supertypes “0632-0636 MH Tac2 Glut”, querying the genes of interest: *Chat*, *Tac1*, *Slc17a7*, *Kcna1-6* and *Oprm1*. *Chat* supertypes were considered those with mean *Chat* expression including zero values > 3 (0632, 0634, and 0635; 6803 cells), while *Tac1* supertypes were defined as those with mean *Tac1* expression including zero values > 3 (0633; 878 cells). Violin and dot plots depicting gene expression levels were created in Graphpad Prism.

Statistical Analyses

Tests for normality (Shapiro-Wilk test) were performed on all datasets to be compared prior to statistical analyses and appropriate parametric (unpaired or paired t-tests) or non-parametric (Mann-Whitney U or Wilcoxon signed rank tests) were accordingly performed. Exact p-values are stated within the figures and values of $p < 0.05$, $p < 0.01$ or $p < 0.001$ were deemed statistically significant and designated with 1, 2 or 3 asterisks, respectively. Error bars in the figures represent standard error of the means unless otherwise indicated. Box whisker plots were constructed as follows: symbol denotes mean value; line represents the median value; lower and upper box limits represent 25th and 75th percentiles, respectively. Lower and upper whiskers represent minimal and maximal data points, respectively.

Data availability

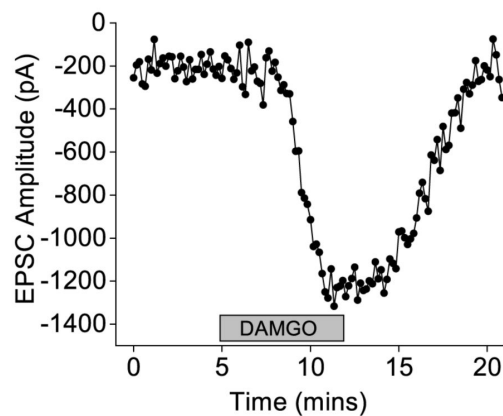
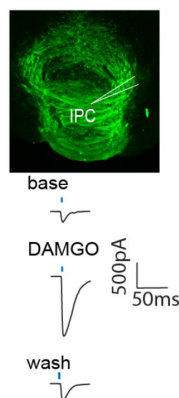
All analyses of data and adapted Python notebooks associated with the manuscript figures are available via the following link (<https://data.mendeley.com/datasets/3dtkmss9xz/1>)

Acknowledgements

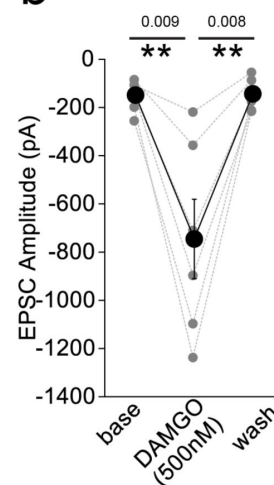
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Additional information

a ChatCre: Ai32 IPC



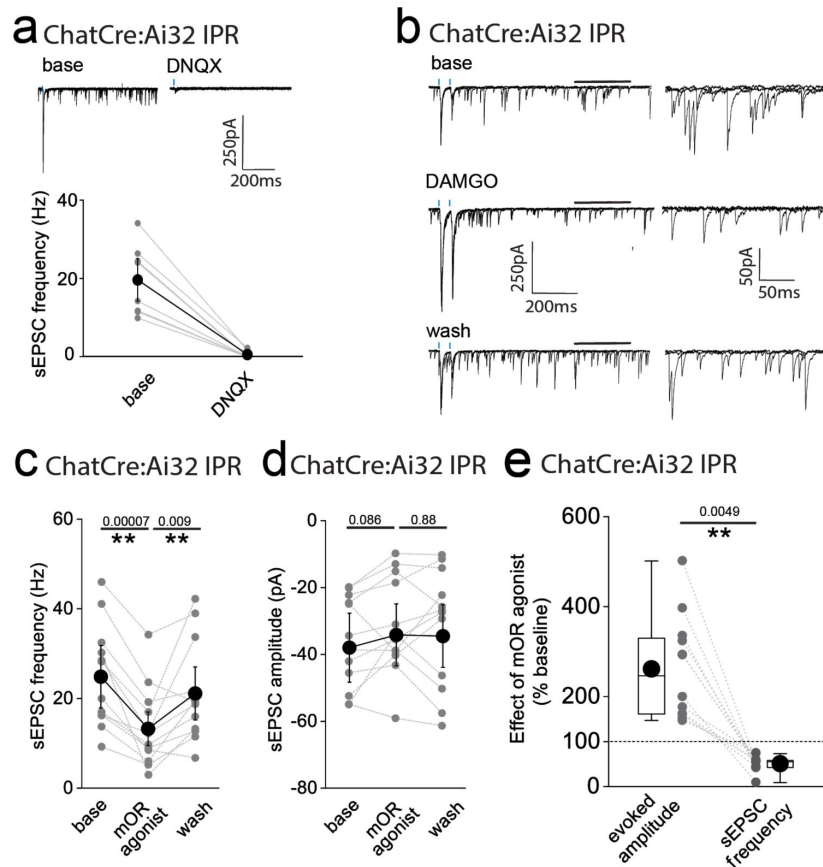
b ChatCre: Ai32 IPC



Supplementary Figure 1.

AMPA-mediated synaptic transmission in central IPN (IPC) mediated by mHB cholinergic neurons is potentiated by MOR activation

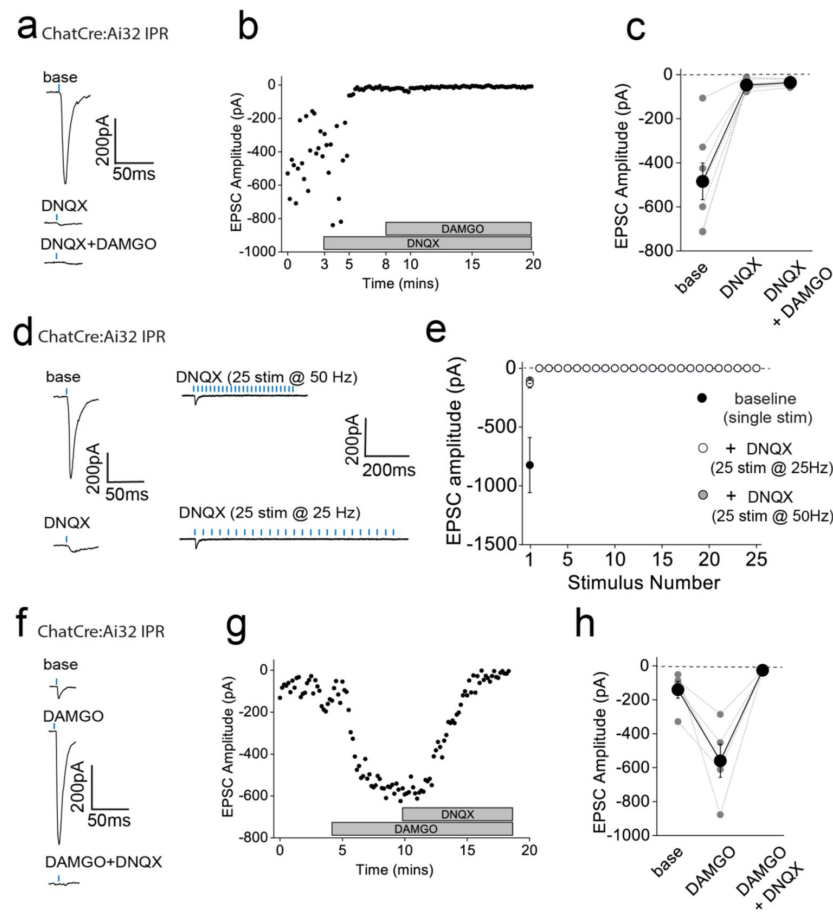
(a) Whole-cell voltage-clamp in adult (>p40) ChatCre: Ai32 mice (top left panel illustrating the axonal arborization of ChR2 expressing cholinergic neuronal axons in IPN and the position of neuronal recording in IPC). Single voltage-clamp traces of light evoked AMPAR EPSCs (bottom left panel; 470nm light pulse; single stimulations, blue dash) and time course of peak amplitudes (right panel) under baseline, during application of 500 nM DAMGO and washout conditions. (b) Individual (grey filled symbols) and mean (black filled symbols) data of AMPAR EPSC amplitude ($n=8$ recorded IPC neurons from 7 mice).



Supplementary Figure 2.

mOR activation imparts opposing effects on spontaneous AMPA EPSCs versus cholinergic neuron mediated evoked AMPAR EPSCs in IPR.

(a) Single voltage-clamp traces (top panels) illustrating light-evoked AMPAR EPSCs mediated by cholinergic neurons and spontaneous EPSCs (sEPSCs) under baseline and following 10 μ M DNQX application in an IPR neuron. Individual (grey symbols) and pooled data (black symbols) illustrating complete cessation of sEPSCs as assessed by frequency following 10 μ M DNQX application ($n = 8$ recorded IPR neurons from 8 mice; bottom panel). (b) Single voltage-clamp traces under baseline, 500nM DAMGO and washout conditions of light-evoked AMPAR EPSCs mediated by mHb cholinergic neurons (2 stimuli at 20Hz; blue dashes) and sEPSCs (left panels). Magnification of sEPSCs events (right panels) corresponding to the region of the traces in the left panels delineated by the black bars. (c,d) Individual (grey symbols) and mean (black symbols) data of sEPSC frequency and amplitude ($n = 12$ recorded IPR cells from 11 mice). (e) Box plot and individual data depicting the relative DAMGO mediated percentage change from baseline of the light-evoked AMPAR EPSC peak amplitude versus AMPAR sEPSC frequency in each individual IPR neuron recorded ($n = 12$ recorded IPR cells from 11 mice).



Supplementary Figure 3.

Effect on DNQX on synaptic transmission mediated by mHB cholinergic neurons onto IPN prior to or after mOR activation and during high frequency stimulation.

(a,b) Voltage-clamp example traces of light-evoked AMPAR EPSCs mediated by cholinergic mHB neurons and time course of peak amplitude under baseline, 10 μ M DNQX and 10 μ M DNQX plus 500nM DAMGO conditions. (c) Individual (grey filled symbols) and mean (black filled symbols) data of AMPAR EPSC amplitude under baseline, 10 μ M DNQX and 10 μ M DNQX plus 500nM DAMGO conditions (n = 7 recorded IPR neurons from 5 mice). (d) Voltage-clamp example traces single stimulus light-evoked AMPAR EPSCs mediated by cholinergic mHB neurons stimulation in the presence and absence of 10 μ M DNQX (left panels). Traces from the same cell in response to trains of light (1ms, 25 stimulations given at 50 or 25Hz; right panels). (e) Pooled data of light evoked responses in response to single stimulation (black) and during 50 and 25 Hz trains (grey and open symbols, respectively; n = 5 recorded cells from 2 mice). (f,g) Voltage-clamp example trace of single light-evoked AMPAR mediated EPSCs mediated by cholinergic mHB neurons and time course of peak amplitude under baseline, 500nM DAMGO and 500nM DAMGO plus 10 μ M DNQX conditions. (d) Individual (grey filled symbols) and mean (black filled symbols) data of AMPAR EPSC amplitude under, baseline, 500nM DAMGO and 500nM DAMGO plus 10 μ M DNQX conditions. (n = 5 recorded IPR neurons from 5 mice).

Author contributions

RC was responsible for project conceptualization, experimental design, electrophysiology experiments, image processing, manuscript writing and figure preparation. CMcB, RC, KAP supervised the project. XY, AV, EL and SH conducted IHC and imaging. AC adapted Python notebooks provided by the Allen Brain Institute to enable analyses and representation of their publicly available mouse whole brain scRNA sequencing data. DA and SH provided critical support for mouse breeding, genotyping and colony maintenance.

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Author information

Ramesh Chittajallu

Section on Cellular and Synaptic Physiology, Eunice Kennedy Shriver National Institute of Child Health and Human Development, National Institutes of Health, Bethesda, United States
ORCID iD: [0000-0002-9794-0052](https://orcid.org/0000-0002-9794-0052)

For correspondence: chittajallu@mail.nih.gov

Anna Vlachos

Section on Cellular and Synaptic Physiology, Eunice Kennedy Shriver National Institute of Child Health and Human Development, National Institutes of Health, Bethesda, United States
ORCID iD: [0000-0002-4411-9447](https://orcid.org/0000-0002-4411-9447)

Adam P Caccavano

Section on Cellular and Synaptic Physiology, Eunice Kennedy Shriver National Institute of Child Health and Human Development, National Institutes of Health, Bethesda, United States
ORCID iD: [0000-0002-6819-533X](https://orcid.org/0000-0002-6819-533X)

Xiaoqing Q Yuan

Section on Cellular and Synaptic Physiology, Eunice Kennedy Shriver National Institute of Child Health and Human Development, National Institutes of Health, Bethesda, United States
ORCID iD: [0000-0003-1737-3295](https://orcid.org/0000-0003-1737-3295)

Steven Hunt

Section on Cellular and Synaptic Physiology, Eunice Kennedy Shriver National Institute of Child Health and Human Development, National Institutes of Health, Bethesda, United States
ORCID iD: [0000-0003-3533-0627](https://orcid.org/0000-0003-3533-0627)

Daniel Abebe

Section on Cellular and Synaptic Physiology, Eunice Kennedy Shriver National Institute of Child Health and Human Development, National Institutes of Health, Bethesda, United States

Edra London

Section on Cellular and Synaptic Physiology, Eunice Kennedy Shriver National Institute of Child Health and Human Development, National Institutes of Health, Bethesda, United States
ORCID iD: [0000-0003-2766-749X](https://orcid.org/0000-0003-2766-749X)

Kenneth A Pelkey

Section on Cellular and Synaptic Physiology, Eunice Kennedy Shriver National Institute of Child Health and Human Development, National Institutes of Health, Bethesda, United States
ORCID iD: [0000-0002-9731-1336](https://orcid.org/0000-0002-9731-1336)

Chris J McBain

Section on Cellular and Synaptic Physiology, Eunice Kennedy Shriver National Institute of Child Health and Human Development, National Institutes of Health, Bethesda, United States
ORCID iD: [0000-0002-5909-0157](https://orcid.org/0000-0002-5909-0157)

Editors

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Reviewer #1 (Public review):

Summary:

In this manuscript, the authors demonstrate for the first time that opioid signaling has opposing effects on the same target neuron depending on the source of the input. Further, the authors provide evidence to support the role of potassium channels in regulating a brake on glutamatergic and cholinergic signaling, with the latter finding being developmentally regulated and responsive to opioid treatment. This evidence solves a conundrum regarding cholinergic signaling in the interpeduncular nucleus that evaded elucidation for many years.

Strengths:

This manuscript provides 3 novel and important findings that significantly advance our understanding of the medial habenula-interpeduncular circuitry:

- (1) Mu opioid receptor activation (mOR) reduces postsynaptic glutamatergic currents elicited from substance P neurons while simultaneously enhancing postsynaptic glutamatergic currents from cholinergic neurons, with the latter being developmentally regulated.
- (2) Substance P neurons from the Mhb provide functional input to the rostral nucleus of the IPN, in addition to the previously characterized lateral nuclei.
- (3) Potassium channels (Kv1.2) provide a brake on neurotransmission in the IPN,

The findings here suggest that the authors have identified a novel mechanism for the normal function of neurotransmission in the IPN, so it would be expected to be observable in almost any animal. In the revised manuscript, the authors put forth significant effort to increase the n, thus increasing the confidence in the observations.

There are also significant sex differences in nAChR expression in the IPN that might not be functionally apparent using the low n presented here. In the revised manuscript, the authors increased the n, and provided data to the reviewers that no significant sex differences were apparent, although there was a trend. Future studies should examine sex differences in detail.

There are also some particularly novel observations that are presented but not followed up on, and this creates a somewhat disjointed story. For example, in Figure 2, the authors identify neurons in which no response is elicited by light stimulation of ChAT-neurons, but application of DAMGO (mOR agonist) un-silences these neurons. Are there baseline differences in the electrophysiological or morphological properties of these "silent" neurons compared to the responsive neurons? In the revised manuscript, the authors directly tested this with new experiments in SST+ neurons in the IPN, demonstrating convincingly that mOR activation un-silences these neurons.

With the revisions, the authors have addressed the reviewers' concerns and significantly improved the manuscript. I find no further weaknesses.

<https://doi.org/10.7554/eLife.106062.2.sa3>

Reviewer #2 (Public review):

Summary:

In this paper, Chittajallu and colleagues present compelling evidence that mu opioid receptor (MOR) activation can potentiate synaptic neurotransmission in a medial habenula to interpeduncular nucleus (mHb-IPN) subcircuit. While, projections from mHb tachykinin 1 (Tac1) neurons onto lateral IPN neurons show a canonical opioid-induced synaptic depression in glutamate release, excitatory neurotransmission in mHb choline acetyltransferase (ChAT) projections to the rostral IPN is potentiated by opioids. This function emerges around age P27 in mice, when MOR expression in the IPN peaks.

Strengths:

Carefully executed electrophysiological experiments with appropriate controls. Interesting description of a neurodevelopmental change in the effects of opioids on mHb-IPN signaling.

Weaknesses:

A minor concern is that the genetic strategy used to target the mHb-IPN pathway (constitutive ChR2 expression in all ChAT+ and Tac1+ neurons) might not specifically target this projection. Future studies are needed to examine the precise mechanism whereby MOR signaling can potentiate glutamatergic neurotransmission in ChAT+ MHb-IPN projections."

<https://doi.org/10.7554/eLife.106062.2.sa2>

Reviewer #3 (Public review):

Summary:

Here the authors describe the role of mORs in synaptic glutamate release from substance P and cholinergic neurons in the medial habenula to interpeduncular nucleus (IPN) circuit in adult mice. They show that mOR activation reduces evoked glutamate release from substance P neurons yet increases evoked glutamate release and Ach release from cholinergic neurons. Unlike glutamate release, Ach release is only detected when potassium channels are blocked with 4-AP or dendrotoxin. The authors also report a previously unidentified glutamatergic input to IPR from SP neurons and describe the developmental timing of mOR- facilitation in adolescent mice.

Strengths:

- The experiments provide new insight into the role of mORs in controlling evoked glutamate release in a circuit with high levels of mORs and established roles in relevant behaviors.
- The experiments are rigorous, and the results are clear cut. The conclusions are supported by the data.
- The findings will be of interest to those working in the field of synaptic transmission and those interested in the function of the medial habenula or interpeduncular nucleus, as well as those seeking to understand the role of opioids on normal and pathological behaviors.

Weaknesses:

- The mechanistic underpinnings of these interesting and novel results are not pursued.

<https://doi.org/10.7554/eLife.106062.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Reviews):

Weaknesses:

Overall I find the data presented compelling, but I feel that the number of observations is quite low (typically n=3-7 neurons, typically one per animal). While I understand that only a few slices can be obtained for the IPN from each animal, the strength of the novel findings would be more convincing with more frequent observations (larger n, more than one per animal). The findings here suggest that the authors have identified a novel mechanism for the normal function of neurotransmission in the IPN, so it would be expected to be observable in almost any animal. Thus, it is not clear to me why the authors investigated so few neurons per slice and chose to combine different treatments into one group (e.g. Figure 2f), even if the treatments have the same expected effect.

This is a well taken suggestion. However, we must point out that we do perform statistical analyses on the original datasets and we believe that our conclusions are justified as acknowledged by the Reviewer. As the Reviewer is aware, the IPN is a small nucleus and with the slicing protocol used, we typically attain 1-2 slices per mouse that are suitable for recordings. Since most of the experiments in the manuscript deals with some form of pharmacological interrogation, we were reticent to use slices that are not naïve and therefore in general did not perform more than 1 cell recording per slice. Having said this, to comply with the Reviewer's suggestion we have now performed additional experiments to increase the n number for certain experiments. We have amended all figures and legends to incorporate the additional data. We must point out that during the replotting of the data in

the summary Figure 8i (previously Figure 7i) we noticed an error with the data representation of the TAC IPL data and have now corrected this oversight

Figure 2b,c.

500nM DAMGO effect on TAC IPL AMPAR EPSC – n increased from 5 to 9

Figure 3g.

500nM DAMGO effect on CHAT IPR AMPAR EPSC – n increased from 8 to 16 Effect of CTAP on DAMGO on CHAT IPR AMPAR EPSC – n increased from 4 to 7

Figure 3i.

500nm DAMGO or Met-enk effect in “silent” CHAT IPR AMPAR EPSC – n increased from 7 to 9

Figure 4e.

500nM DAMGO effect on ES coupling – Note: in the original version the n number was 5 and not 7 as written in the figure legend. We have now increased the n from 5 – 9.

Figure 5e,f.

500nM DAMGO effect on TAC IPR AMPAR EPSC – n increased from 5 to 9

Figure 7f.

Effect of DH β E on EPSC amplitude after application of DNQX/APV/4-AP or DTX- α – n increased from 7-9.

Figure 7g.

Emergence of nAChR EPSC after DTX – n increased from 4 to 7

Figure 7i.

Effect of ambenonium on nAChR amplitude and charge – n increased from 4 to 7

Supplementary Figure 3c and h

Effect of DAMGO after DNQX – n increased from 4 to 7

Effect of DNQX after DAMGO mediated potentiation – n increased from 3 to 5.

Throughout the study (Figs. 3i, 7f and 8h in the revised manuscript) we do indeed pool datasets that were amassed from different conditions since we were not directly investigating the possibility of any deviation in the extent of response between said treatments. For example, and as pointed out by the Reviewer, in Fig. 2F (now Fig. 3i) the use of DAMGO and met-ENK were merely employed to ascertain whether light-evoked synaptic transmission (ChATCre:ai32 mice) in cells that had no measurable EPSC could be pharmacologically “unsilenced” by mOR activation. Thus, the means by which mOR receptor was activated was not relevant to this specific question. Note: 2 more recordings are now added to this dataset (Fig. 3i) that were taken from ChATChR2/SSTCre:ai9 mice in response to the comment by this Reviewer below (“Are there baseline differences in the electrophysiological or morphological properties of these “silent” neurons compared to the responsive neurons?”). Similarly, in the revised Fig. 7f we pooled data investigating the pharmacological block of the EPSC that emerged following application of either DNQX/APV/4-AP or DNQX/APV/DTX. Low concentrations 4-AP or DTX were interchangeably employed to reveal the DNQX-insensitive EPSC that we go on to show is indeed the nAChR response. Finally, in Fig. 8h, we pooled data demonstrating a lack of effect of DAMGO in potentiating both the glutamatergic and

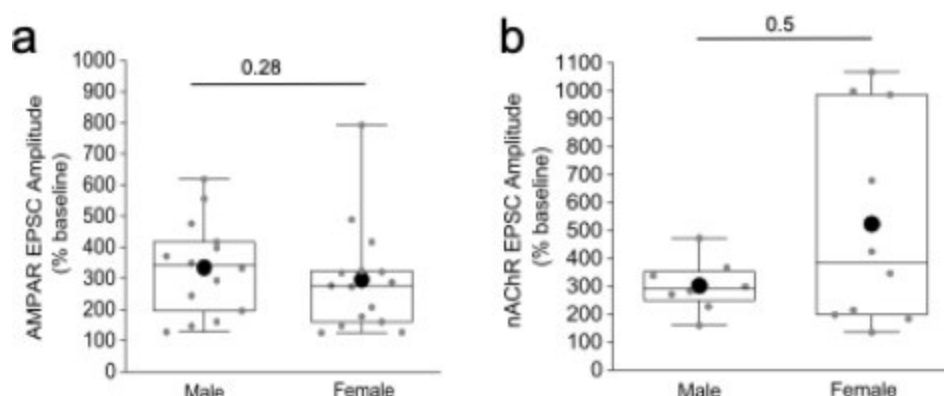
cholinergic arms of synaptic transmission in the OPRM1 KO mice. Again, here we were only interested in determining whether removal of mOR expression prevented potentiation of transmission mediated by mHB ChAT neurons irrespective of neurotransmitter modality. Thus, overall we were careful to only pool data in those instances where it would not change the interpretation and hence conclusions reached.

There are also significant sex differences in nAChR expression in the IPN that might not be functionally apparent using the low n presented here. It would be helpful to know which of the recorded neurons came from each sex, rather than presenting only the pooled data.

As the reviewer correctly states there are veins of literature concerning a divergence, based on sex, of not only nicotinic receptor expression but also behaviors associated with nicotine addiction. However, we have reanalyzed our datasets focusing on the extent of the mOR potentiation of glutamatergic and cholinergic transmission mediated by mHB ChAT neurons in IPR between male and female mice. Please refer to the Author response image 1 below. Although there is a possible trend towards a higher potentiation of nAChR in female mice, this was not found to be of statistical significance (see Author response image 1 below). We therefore chose not to split our data in the manuscript based on gender.

Author response image 1.

Comparison of the mOR (500nM DAMGO) mediated potentiation on evoked (a) AMPAR and (b) nAChR EPSCs in IPR between male and female mice.



There are also some particularly novel observations that are presented but not followed up on, and this creates a somewhat disjointed story. For example, in Figure 2, the authors identify neurons in which no response is elicited by light stimulation of ChAT-neurons, but the application of DAMGO (mOR agonist) un-silences these neurons. Are there baseline differences in the electrophysiological or morphological properties of these "silent" neurons compared to the responsive neurons?

Unfortunately, we did not routinely measure intrinsic properties of the recorded postsynaptic neurons nor systematically recovered biocytin fills to assess morphology. Therefore, it remains unclear whether the neurons in which there were none or minimal AMPAR-mediated EPSCs are distinct to the ones displaying measurable responses. The IPR is resident to GABAergic SST neurons that comprise the most numerous neuron type in this IPN subdivision. Although heavily outnumbered by the SST neurons there are additionally VGlut3+ glutamatergic neurons in IPN. The Reviewer is likely referring to a recent study investigating synaptic transmission specifically onto SST+ and VGlut3+ neurons in IPN demonstrating that mHB cholinergic mediated glutamatergic input is "weaker" onto the

glutamatergic neurons. Furthermore, in some instances synaptic transmission onto this latter population can be “unsilenced” by GABAB receptor activation in a similar manner to that seen with mOR activation in this manuscript when IPR neurons are blindly targeted (Stinson & Ninan, 2025). Using a similar strategy as in this recent study (Stinson & Ninan, 2025), we now include experiments in which the ChATChR2 mouse was crossed with a SSTCre: Ai14. This allowed for recording of postsynaptic EPSCs in directly identified SST IPR neurons. We demonstrate that DAMGO can indeed increase glutamatergic EPSCs and in 2 of the cells where light activation demonstrated no appreciable AMPAR EPSC upon maximal LED light activation, DAMGO clearly “unsilenced” transmission. Thus, our additional analyses directly demonstrate that our original observations concerning mOR modulation extend to the mHb cholinergic AMPAR mediated input onto IPR SST neurons. This additional data is in the revised manuscript (Figure 3D-F, I). Future experimentation will be required to determine if the propensity of encountering a “silent” input that can be converted to robust synaptic transmission by mOR differs between these two cell types. Furthermore, it will be of interest to investigate if any differences exist in the magnitude of the cholinergic input or the mOR mediated potentiation of co-transmission between postsynaptic SST GABA and glutamatergic neuronal subtypes.

Reviewer #2 (Public review)

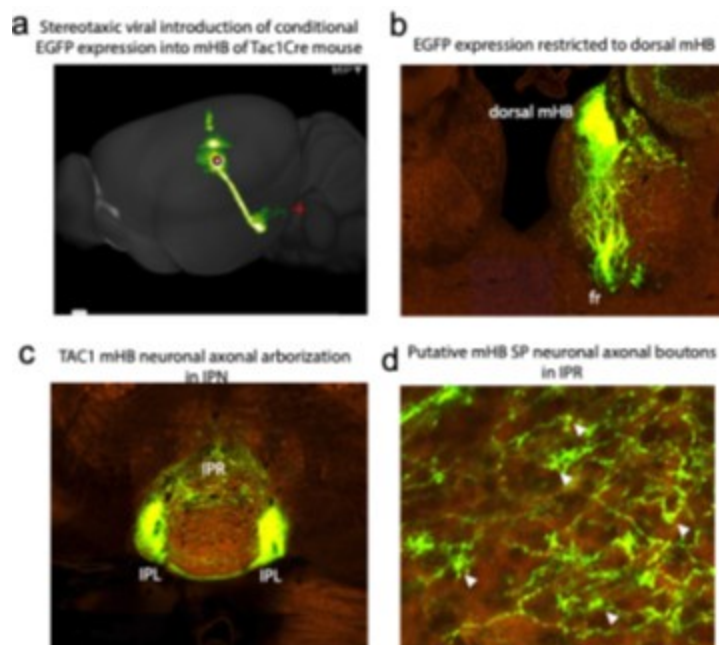
Weaknesses:

The genetic strategy used to target the mHb-IPN pathway (constitutive expression in all ChAT+ and Tac1+ neurons) is not specific to this projection.

This is an important point made. We are acutely aware that the source of the synaptic input in IPN mediated by conditional expression of ChR2 employing using transgenic cre driver lines does not confer specificity to mHb. This is particularly relevant considering one of the novel observations here relates to a previously unidentified functional input from TAC1 neurons to the IPR. At this juncture we would like to point the Reviewer to the publicly available Connectivity Atlas provided by the Allen Brain Institute (<https://connectivity.brain-map.org/>). With reference to mHb TAC1 neuronal output, targeted viral injection into the habenula of Tac1Cre mice allows conditional expression of EGFP to SP neurons as evidenced by the predominant expression of reported fluorescence in dorsal mHb (see Author response image 2 a,b below). Tracing the axonal projections to the IPN clearly demonstrates dense fibers in IPL as expected but also arborization in IPR (Author response image 2 a,c). This pattern is reminiscent of that seen in the transgenic Tac1Cre: ai9 or ai32 mice used in the current study (Figs. 1c, 2a, 5c). Closer inspection of the fibers in the IPR reveals putative synaptic bouton like structures as we have shown in Fig. 5a,b (Author response image 2 d below).

Author response image 2.

Sterotaxic viral injection into mHb pf Tac1Cre mice taken from Allen Brain connectivity atlas (Link to Connectivity Atlas for mHb SP neuronal projection pattern)



These anatomical data suggest that part of the synaptic input to the IPR originates from mHB TAC1 neurons although we cannot fully discount additional synaptic input from other brain areas that may impinge on the IPR. Indeed, as the Reviewer points out, it is evident that other regions including the nucleus incertus send outputs to the IPN (Bueno et al., 2019; Liang et al., 2024; Lima et al., 2017). However, it is unclear if neuronal inputs from these alternate sources {Liang, 2024 #123; Lima, 2017 #33} {Bueno, 2019 #178} are glutamatergic in nature AND mediated by a TAC1/OPRM1-expressing neuronal population. Nevertheless, we have now modified text in the discussion to highlight the limitations of using a transgenic strategy (pg 12, para 1).

In addition, a braking mechanism involving Kv1.2 has not been identified.

It is unclear to what the Reviewer is referring to here. Although most of our experiments pertaining to the brake on cholinergic transmission by potassium channels use low concentrations of 4-AP (50100 μ M) which have been used to block Shaker Kv1 channels there although at these concentrations there are additional action at other K⁺-channels such as Kv3, for instance. However, we essentially demonstrate that a selective Kv1.1 and Kv1.2 antagonist dendrotoxin replicates the 4-AP effects. We have now also included RNAseq data demonstrating the relative expression levels of Kv1 channel mRNA in mHb ChAT neurons (KCNA1 through KCNA6; Figure 6b). The complete absence of KCNA1 yet a high expression level of KCNA2 transcripts highly suggests a central role of Kv1.2 in unmasking nAChR mediated synaptic transmission.

Reviewer #3 (Public review)

Weaknesses:

The significance of the ratio of AMPA versus nACh EPSCs shown in Figure 6 is unclear since nAChR EPSCs measured in the K⁺ channel blockers are compared to AMPA EPSCs in control (presumably 4-AP would also increase AMPA EPSCs).

We understand the Reviewer's concern regarding the calculation of nicotinic/AMPA ratios since they are measured under differing conditions i.e. absence and presence of 4-AP,

respectively. As the reviewer correctly points out 4-AP likely increases the amplitude of the AMPA receptor mediated EPSC. However, our intention of calculating this ratio was not to ascertain a measure of relative strengths of fast glutamatergic vs cholinergic transmission onto a given postsynaptic IPN neuron per se. Rather, we used the ratio as a means to normalize the size of the nicotinic receptor EPSC to the strength of the light stimulation (using the AMPA EPSC as the normalizing factor) in each individual recording. This permits a more meaningful comparison across cells/slices/mice. We apologize for the confusion and have amended the text in the results section to reflect this (pg 9; para2).

The mechanistic underpinnings of the most novel results are not pursued. For example, the experiments do not provide new insight into the differential effects of evoked and spontaneous glutamate/ACh release by Gi/o coupled mORs, nor the differential threshold for glutamate versus ACh release.

Our major goal of the current manuscript was to provide a much-needed roadmap outlining the effects of opioids in the habenulo-interpeduncular axis. Of course, a full understanding of the mechanisms underlying such complex opioid actions at the molecular level will be of great value. We feel that this is beyond the scope of this already quite result dense manuscript but will be essential if directed manipulation of the circuit is to be leveraged to alter maladaptive behaviors associated with addiction/emotion during adolescence and in adult.

The authors note that blocking Kv1 channels typically enhances transmitter release by slowing action potential repolarization. The idea that Kv1 channels serve as a brake for ACh release in this system would be strengthened by showing that these channels are the target of neuromodulators or that they contribute to activity-dependent regulation that allows the brake to be released.

The exact mechanistic underpinnings that can potentially titrate Kv1.2 availability and hence nAChR transmission would be essential to shed light on potential in vivo conditions under which this arm of neurotransmission can be modulated. However, we feel that detailed mechanistic interrogation constitutes significant work but one that future studies should aim to achieve. Thus, it presently remains unclear under what physiological or pathological scenarios result in attenuation of Kv1.2 to subsequently promote nAChR mediated transmission but as mentioned in the existing discussion future work to decipher such mechanisms would be of great value.

Reviewer #1 (Recommendations for the authors):

Overall I find this to be a very interesting and exciting paper, presenting novel findings that provide clarity for a problem that has persisted in the IPN field: that of the conundrum that light-evoked cholinergic signaling was challenging to observe despite the abundance of nAChRs in the IPN.

Major concerns:

(1) The n is quite low in most cases, and in many instances, data from one figure are replotted in another figure. Given that the findings presented here are expected in the normal condition, it should not be difficult to increase the n. A more robust number of observations would strengthen the novel findings presented here.

Please refer to the response to the public review above.

(2) In general, I find the organization of the figures somewhat disjointed. Sometimes it feels as if parts of the information presented in the results are split between figures,

where it would make more sense to be together in a figure. For example, all the histology for each of the lines is in Figure 1, but only ephys data for one line is included there. It would be more logical to include the histology and ephys data for each line in its own figure. It would also be helpful to show the overlap of mOR expression with Tac1-Cre and ChAT-Cre terminals in the IPN. Likewise, the summarized Tac1Cre: Ai32 IPR data is in Figure 4, but the individual data is in Figure 5.

We introduce both ChAT and TAC1 cre lines in Figure 1 as an overview particularly for those readers who are not entirely familiar with the distinct afferent systems operating with the habenulointerpeduncular pathway. However, in compliance with the Reviewer's suggestion we have now restructured the Figures. In the revised manuscript, the functional data pertaining to the various transmission modalities mediated by the distinct afferent systems impinging on the subdivision of the IPN tested are now split into their own dedicated figure as follows:

Figure 2.

mOR effect on TAC1 neuronal glutamatergic output in IPL.

Figure 3.

mOR effect on CHAT neuronal glutamatergic output in IPR.

Figure 5.

mOR effect on TAC1 neuronal glutamatergic output in IPR.

Figure 8.

mOR effect on CHAT neuronal cholinergic output in IPC.

Supp. Fig. 1 mOR effect on CHAT neuronal glutamatergic output in IPC.

We thank the Reviewer for their suggestions regarding the style of the manuscript. The restructuring has now resulted in a much better flow of the presented data.

(3) The discussion is largely satisfactory. However, a little more discussion of the integrative function of the IPN is warranted given the opposing effects of MOR activation in the Tac vs ChAT terminals, particularly in the context of both opioids and natural rewards.

We thank the reviewer for this comment. However, we feel the discussion is rather lengthy as is and therefore we refrained from including additional text.

Minor concerns:

(1) The methods are missing key details. For example, the stock numbers of each of the strains of mice appear to have been left out. This is of particular importance for this paper as there are key differences between the ChAT-Cre lines that are available that would affect observed electrophysiological properties. As the authors indicate, the ChAT-ChR2 mice overexpress VACHT, while the ChAT-IRES-Cre mice do not have this problem. However, as presented it is unclear which mice are being used.

We apologize for the omission - the catalog numbers of the mice employed have now been included in the methods section.

We have now clearly included in each figure panel (single trace examples and pooled data) from which mice the data are taken from – in some instances the pooled data are from the

two CHAT mouse strains employed. Despite the tendency of the ChATChR2 mice to demonstrate more pronounced nAChR mediated transmission (Fig. 7h), we justify pooling the data since we see no statistical significance in the effect of mOR activation on either potentiating AMPA or nAChR EPSCs (Please refer to response to Reviewer 2, Minor Concern point 2)

(2) Likewise, antibody dilutions used for staining are presented as both dilution and concentration, which is not typical.

We thank the reviewer for pointing out this inconsistency. We have amended the text in the methods to include only the working dilution for all antibodies employed in the study.

(3) There are minor typos throughout the manuscript.

All typos have been corrected.

Reviewer #2 (Recommendations for the authors):

The authors provide a thorough investigation into the subregion, and cell-type effect of mu opioid receptor (MOR) signaling on neurotransmission in the medial habenula to interpeduncular nucleus circuit (mHb-IPN). This circuit largely comprises two distinct populations of neurons: mHb substance P (Tac1+) and cholinergic (ChAT+) neurons. Corroborating prior work, the authors report that Tac1+ neurons preferentially innervate the lateral IPN (IPL) and rostral IPN (IPR), while ChAT+ neurons preferentially innervate the central IPN (IPC) and IPR. The densest expression of MOR is observed in the IPL and MOR agonists produce a canonical presynaptic depression of glutamatergic neurotransmission in this region. Interestingly, MOR signaling in the ChAT+ mHb projection to the IPR potentiates light-evoked glutamate and acetylcholine-mediated currents (EPSC), and this effect is mediated by a MOR-induced inhibition of Kv2.1 channels.

Major concerns:

(1) The method used for expressing channelrhodopsin (ChR2) into cholinergic and neurokinin neurons in the mHb (Ai32 mice crossed with Cre-driver lines) has limitations because all Tac1+/ChAT+ inputs to the IPN express ChR2 in this mouse. Importantly, the IPN receives inputs from multiple brain regions besides the IPN-containing neurons capable of releasing these neurotransmitters (PMID: 39270652). Thus, it would be important to isolate the contributions of the mHb-IPN pathway using virally expressed ChR2 in the mHb of Cre driver mice.

Please refer to the response to the public review above.

(2) Figure 4: The authors conclude that the sEPSC recorded from IPR originate from Tac1+ mHbIPR projections. However, this cannot be stated conclusively without additional experimentation. For instance, an optogenetic asynchronous release experiment. For these experiments it would also be important to express ChR2 virus in the mHb in Tac1- and ChAT-Cre mice since glutamate originating from other brain regions could contribute to a change in asynchronous EPSCs induced by DAMGO.

This is a well taken point. The incongruent effect of DAMGO on evoked CHAT neuronal EPSC amplitude and sEPSC frequency prompted us to consider the the possibility of differing effect of DAMGO on a secondary input. We agree that we do not show directly if the sEPSCs originate from a TAC1 neuronal population. Therefore, we have tempered our wording with

regards the origin of the sEPSCs and have also restructured the Figure in question moving the sEPSC data into supplemental data (Supplemental Fig. 2)

(3) Figure 5D: It would be useful to provide a quantitative measure in a few mice of mOR fluorescence across development (e.g. integrated density of fluorescence in IPR).

We have now included mOR expression density across development (Fig. 6). Interestingly, the adult expression levels of mOR in the IPR are essentially reached at a very early developmental age (P10) yet we see stark differences in the role of mOR activation in modulating glutamatergic transmission mediated by mHB cholinergic neurons. Note: since we processed adult tissue (i.e. >p40) for these developmental analyses we utilized these slices to also include an analysis of the relative mOR expression density specifically in adults between the subdivisions of IPN in Fig. 1.

(4) Figure 6B: It would be useful to quantify the expression of Kcna2 in ChAT and Tac1 neurons (e.g. using FISH).

We thank the Reviewer for this suggestion. We have now included mRNA expression levels available from publicly available 10X RNA sequencing dataset provided by the Allen Brain Institute (Figure 7b).

(5) It would be informative to examine what the effects of MOR activation are on mHb projections to the (central) .

In response to this suggestion, we now have included additional data in the manuscript in putative IPC cells that clearly demonstrate a similar DAMGO elicited potentiation of AMPAR EPSC to that seen in IPR. These data are now included in the revised manuscript (Supplemental Fig. 1; Fig. 8i).

(6) What is the proposed link between MOR activation and the inhibition of Kv1.2 (e.g. beta-Arrestin signaling, G beta-gamma interaction with Kv1.2, PKA inhibition?)

We apologize for any confusion. We do not directly test whether the potentiation of EPSCs upon mOR activation occurs via inhibition of Kv1.2. Although we have not directly tested this possibility we find it an unlikely underlying cellular mechanism, especially for the potentiation of the cholinergic arm of neurotransmission since in the presence of DNQX/APV, the activation of mOR does not result in any emergence of any nAChR EPSC (see Supplementary Fig. 3a-c)

Minor concerns:

(1) Methods: Jackson lab ID# for used mouse strains is missing.

We apologize for this omission and have now included the mouse strain catalog numbers.

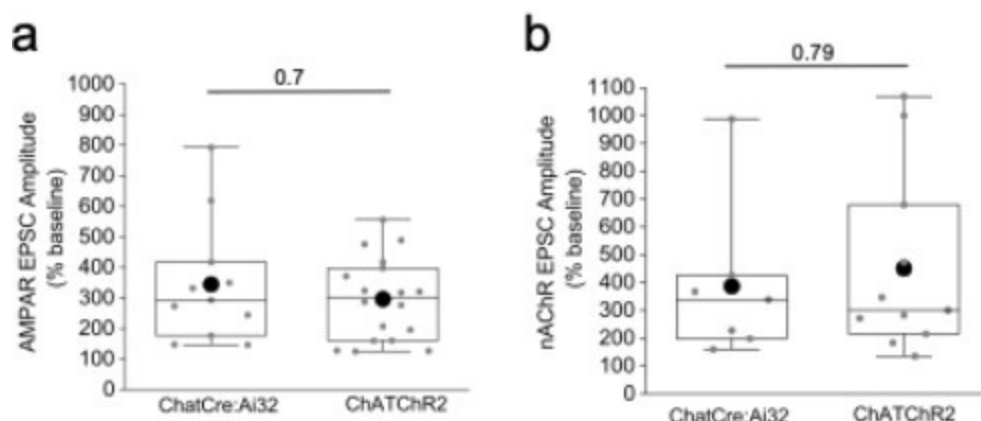
(2) The authors use data from both ChAT-Cre x Ai32 and ChAT-ChR2 mice. It would be helpful to show some comparisons between the lines to justify merging data sets for some of the analyses as there appear to be differences between the lines (e.g. Figure 6G).

This is a well taken point. We have now provided a figure for the Reviewer (see below) that illustrates the lack of significant difference between the mOR mediated potentiation of both mHB CHAT neuronal AMPAR and nAChR transmission between the two mouse lines employed despite a divergence in the extent of glutamatergic vs cholinergic transmission

shown in Fig. 7g (previously Figure 6g). We have chosen not to include this data in the revised manuscript.

Author response image 3.

Comparison of the mOR (500nM DAMGO) mediated potentiation on evoked AMPAR **(a)** and nAChR **(b)**EPSCs in IPR between ChatCre: Ai32 and ChatChR2 mice.



(3) Line 154: How was it determined that the EPSC is glutamatergic?

We apologize for any confusion. In the revised manuscript we now clearly point to the relevant figures (see Supplementary Figs. 2a and 3) in the Results section (pg. 4, para 2; pg 7, para 1; pg 8, para2) where we determine that both the sEPSCs and ChAT mediated light evoked EPSCs recorded under baseline conditions are totally blocked by DNQX and hence are exclusively AMPAR events

(4) It would be helpful to discuss the differences between GABA-B mediated potentiation of mHbIPN signaling and the current data in more detail.

We are unclear as to what differences the Reviewer is referring to. At least from the perspective of ChAT neuronal mediated synaptic transmission, other groups (and in the current study; **Fig. 7h**) have clearly shown that GABA_B activation markedly potentiates synaptic transmission like mOR activation. Nevertheless, based on our novel findings it would be of interest to determine whether the influence of GABA_B is inhibitory onto the TAC mediated input in IPR and whether there is a developmental regulation of this effect as we demonstrate upon mOR activation. These additional comparisons between the effect of the two Gi-linked receptors may shed light onto the similarity, or lack thereof, regarding the underlying cellular mechanisms. We now have included a few sentences in the discussion to highlight this (pg 11, para 1).

Reviewer #3 (Recommendations for the authors):

The abstract was confusing at first read due to the complex language, particularly the sentence starting with... Further, specific potassium channels...

The authors might want to consider simplifying the description of the experiments and the results to clarify the content of the manuscript for readers who many only read the abstract.

We have altered the wording of the abstract and hope it is now more reader friendly.

The opposite effect of mOR activation on spontaneous EPSCs versus electrical or ChR2-evoked EPSCs is very interesting and raises the issue of which measure is most physiologically relevant. For example, it is unclear whether sEPSCs arise primarily from cholinergic neurons (that are spontaneously active in the slice, Figure 3), and if so, does mOR activation suppress or enhance cholinergic neuron excitability and/or recruitment by ChR2? While a full analysis of this question is beyond the scope of this manuscript, the assumption that glutamate release assayed by electrical/ChR2 evoked transmission is the most physiologically relevant might merit some discussion since sEPSCs presumably also reflect action-potential dependent glutamate release. One wonders whether mORs hyperpolarize cholinergic neurons to reduce spontaneous spiking yet enhance fiber recruitment by ChR2 or an electrical stimulus (i.e. by removing Na channel inactivation). The authors have clearly stated that they do not know where the mORs are located, and that the effects arising from disinhibition are likely complex. But they also might discuss whether glutamate release following synchronous activation of a fiber pathway by ChR2 or electrode is more or less physiologically relevant than glutamate release assayed during spontaneous activity. It seems likely that an equivalent experiment to Figure 3D, E using spontaneous spiking of IPR neurons would show that spiking is reduced by mOR activation.

We thank the Reviewer for this comment. As pointed it would be of interest to dissect the “network” effect of mOR activation but as the Reviewer acknowledges this is beyond the scope of the current manuscript. The Reviewer is correct in postulating that mOR activation results in hyperpolarization of mHB ChAT neurons. A recent study(Singhal et al 2025) demonstrate that a subpopulation of ChAT neurons undergoes a reduction in firing frequency following DAMGO application. This is corroborated by our own observations although we chose not to include this data in our current manuscript (but see below).

Additionally, the Reviewer questions whether ChR2/electrical stimulation is physiological. This is a well taken point and of course the simultaneous activation of potentially all possible axonal release sites is not the mode under which the circuit operates. Nevertheless, our data clearly demonstrates the ability of mORs to modulate release under these circumstances that must reflect an impact on spontaneous action potential driven evoked release. Although the suggested experiment could shed light on the synaptic outcomes of mOR receptor activation on ES coupling of downstream IPN neurons. Interpretation of the outcome would be confounded by the fact that postsynaptic IPN neurons also express mORs . Thus, we would not be able to isolate the effects of presynaptic changes in modulating ES coupling from any direct postsynaptic effect on the recorded cell when in current clamp.

Together these additional sites of action of mOR (i.e. mHB ChAT somatodendritic and postsynaptic IPN neuron) only serve to further highlight the complex nature of the actions of opioids on the habenulo-interpeduncular axis warranting future work to fully understand the physiological and pathological effects on the habenulo-interpeduncular axis as a whole.

The idea that Kv2.1 channels serve as a brake raises the question of whether they contribute to activity-dependent action potential broadening to facilitate Ach release during trains of stimuli.

This is an interesting suggestion and one that we had considered ourselves. Indeed, as the Reviewer is likely aware and as mentioned in the manuscript, previous studies have shown nAChR signaling can be revealed under conditions of multiple stimulations given at relatively high frequencies. We therefore attempted to perform high frequency stimulation (20 stimulations at 25Hz and 50Hz) in the presence of ionotropic glutamatergic receptor antagonists DNQX and APV. We have now included this data in the revised manuscript (Supplementary Fig 3b). As shown, this failed to engage nAChR mediated synaptic

transmission in our hands. Interestingly there is evidence from reduced expression systems demonstrating that Kv1.2 channels undergo use-dependent potentiation (Baronas et al., 2015) in contrast to that seen with other K⁺-channels. Whether this is the case for the axonal Kv1.2 channels on mHB axonal terminals in situ is not known but this may explain the inability to reveal nAChR EPSCs upon delivery of such stimulation paradigms.

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